KGDNet: Knowledge-Guided Decoupled Modeling of Heterogeneous Change Mechanisms for Remote Sensing Change Detection

Zongyu Yang, Haitao Li, Haiyan Gu, Yi Yang, Shaopeng Ding, and Haozhu Kong

***Abstract*—High-resolution remote sensing change detection (RSCD) methods typically rely on single-modal image features, making them highly susceptible to pseudo-changes induced by factors like illumination, shadows, and seasonal variations. Moreover, most approaches adopt a unified binary change paradigm, overlooking the heterogeneous mechanisms of change, such as appearance and disappearance. To address these limitations, this article presents KGDNet, a knowledge-guided tri-branch network designed for decoupled modeling of heterogeneous change mechanisms in change detection (CD). KGDNet leverages annotation-free monocular depth maps as geometric priors and, through the guided alignment and fusion module (GAFM), upgrades 2-D features to 3-D geometry-aware representations. This enhances the discrimination of genuine land-cover changes while effectively suppressing pseudo-changes at the source. The network explicitly decouples three core mechanisms—appearance, difference, and disappearance—using specialized parallel branches to capture distinct change patterns. Additionally, the difference branch dynamically corrects and refines prior knowledge, mitigating contamination from low-quality depth cues. To efficiently process high-resolution imagery, we further propose a multihead latent spatiotemporal attention (MH-LSTA) module that compresses and remaps features in a latent space, substantially reducing computational and memory costs while enhancing the modeling of global spatiotemporal relations. This yields a favorable tradeoff between accuracy and efficiency. Comprehensive experiments on public benchmarks demonstrate that KGDNet outperforms representative state-of-the-art (SOTA) methods in key metrics, delivering superior detection accuracy and robustness. This study proposes a knowledge-guided decoupled modeling paradigm that integrates prior knowledge with heterogeneous change mechanisms to enhance the interpretability of change representations. The code and data will be released upon publication at https://github.com/yangzy1217/KGDNet**

***Index Terms*—Change detection (CD), decoupled change modeling, knowledge-guided learning, multimodal data alignment, remote sensing.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to automatically identify spatiotemporal differences in land-cover states from multitemporal images and is fundamental to national land monitoring, urban expansion assessment, disaster damage interpretation, and environmental supervision [1], [2]. With Earth observation entering a new era of high-resolution, high revisit frequency, and multisource sensing, the wealth of image details creates unprecedented opportunities to move beyond the binary question of whether changes occur and toward a deeper mechanistic understanding of how they occur [3]. However, this progress also amplifies pseudo-changes arising from scale inconsistencies, illumination and shadow variations, seasonal phenology, sensor viewpoint differences, and subpixel misregistration [4], [5], [6]. As illustrated in Fig. 1, even in the absence of real land-cover changes, significant appearance discrepancies can be observed due to illumination and shadow variations, vegetation phenology, and large intraclass variations in building textures and colors. Meanwhile, ever-increasing image resolution and swath widths strain conventional global attention and heavy spatiotemporal modeling, resulting in computational and memory bottlenecks [7]. Consequently, a central challenge is to reliably separate genuine land-cover changes from pseudo-changes under a controllable computational budget.

Most existing approaches are dominated by 2-D appearance features, whose accuracy is susceptible to imaging conditions and data quality [8], [9], [10], [11]. Although some studies introduce priors, such as digital elevation models (DEMs) [12], OpenStreetMap (OSM) [13], or edge maps [14], [15], to constrain learning, these solutions are hindered by the availability of external data and the high cost of annotation, which limits scalability. Recently, the emergence of general monocular depth estimation (MDE) offers a promising alternative; high-quality depth maps can be generated from single optical images without extra labels [16], [17], [18], alleviating the dependence on curated priors and enabling robust geometric constraints for scale normalization and view alignment in end-to-end networks. While some pioneering studies have recently attempted to incorporate such depth priors into RSCD, they typically adopt a task-level coupling paradigm, such as multitask feature fusion or pseudo-label distillation. These approaches implicitly treat the generated depth maps as

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Fig. 1. Examples of pseudo-changes in bitemporal remote sensing images. (a) T1 image (red boxes). (b) T2 image (blue boxes). The zoomed-in patches illustrate specific challenges: 1) pseudo-changes caused by illumination and shadows; 2) pseudo-changes caused by phenology; 3) pseudo-changes in buildings due to texture; and 4) pseudo-changes in buildings due to spectral variations.

noise-free ground truth and forcefully couple them with internal learning objectives. Consequently, they are highly susceptible to being misled by MDE-induced artifacts and often incur substantial computational overhead due to heavy architectures. This shifts the focus to two more fundamental questions. First, how can we transform prior knowledge from being passively concatenated as extra channels to flexible, effective guidance that steers representation learning? Second, how can we identify and suppress erroneous responses within the prior knowledge itself to avoid contaminating model training?

Another limitation of mainstream methods lies in their unified binary change supervision paradigm, which emphasizes the mere presence of change while overlooking the heterogeneous mechanisms underlying different change types [19], [20], [21]. Predicting diverse mechanisms, such as appearance, difference, and disappearance, with a single feature map hinders the full exploitation of their distinct patterns [22]: focusing on newly added objects can lead to missed detections of disappearing targets, while relying solely on absolute differencing is vulnerable to pseudo-change signals [23]. It is thus necessary to decouple change modeling along mechanistic dimensions to analyze land-cover evolution from multiple perspectives [24], thereby capturing complex boundaries more effectively, suppressing pseudo-changes, and improving small-object recognition.

To address these issues, we propose KGDNet: a knowledge-guided network for decoupled change mechanism modeling in RSCD. Unlike binary supervision or boundary-assisted designs, KGDNet explicitly decouples the three core mechanisms and codesigns them with a prior-refinement pathway. Monocular depth maps, obtained from a depth estimation model, are injected as knowledge into the feature learning process to upgrade conventional 2-D appearance features to 3-D geometry-aware representations. By introducing relative height, spatial connectivity, and regional consistency cues at the pixel, geometric, and structure levels [3], the framework drives geometric alignment and scale normalization. As a result, pseudo-changes induced by phenology, illumination, and shadows are substantially suppressed. For high-resolution imagery, we further develop a multihead latent spatiotemporal attention (MH-LSTA) module with lightweight geometry-aware positional encoding (Geo-PE). Features are compressed and remapped in a latent space to reduce memory and computation while strengthening geometric structure and spatial distribution modeling, achieving stable gains with minimal parameter overhead. To fit different mechanisms, a tri-branch collaborative architecture is adopted. The appearance branch focuses on newly emerging objects, the disappearance branch models object degradation and vanishing, and the difference branch learns explicit cross-temporal differencing while dynamically refining the priors to suppress nontarget responses, such as trees and shadows. The three branches learn in parallel across multiple scales and are adaptively fused at the end to enable a fine-grained interpretation of change types. The main contributions of this work are summarized as follows.

1) We propose KGDNet, a decoupled multimodal fusion paradigm for RSCD. Unlike the conventional multitask paradigms that tightly couple depth extraction as a parallel learning target, KGDNet utilizes MDE-derived depth maps as flexible extrinsic guidance. By injecting these label-free depth maps as knowledge into end-to-end learning, it lifts 2-D appearance features into 3-D geometry-aware representations. Under multilevel constraints at the pixel, geometric, and structure levels, KGDNet suppresses pseudo-changes and better captures small objects and complex boundaries, achieving state-of-the-art (SOTA) performance on multiple benchmarks.
2) To effectively exploit multidimensional features while reducing redundant learning and parameter counts across modules, we propose a unified, weight-sharing guided alignment and fusion module (GAFM) that performs knowledge-guided cross-modal alignment and cross-temporal fusion. Moreover, because processing high-resolution remote sensing imagery imposes greater computational and memory demands, we design an MH-LSTA block within GAFM. It leverages positional encoding together with latent-space compression and remapping to enhance geometric awareness. This further reduces memory footprint and computational cost without sacrificing accuracy.
3) The existing RSCD paradigm focuses on training a single network to simultaneously learn heterogeneous change mechanisms, which limits the model's understanding of the underlying change dynamics. To overcome this limitation, we propose a mechanism-decoupled modeling approach. Instead of learning a single, undifferentiated change signal, we explicitly model appearance and disappearance in parallel, while

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introducing a third, critical difference branch. Crucially, this branch actively evaluates and refines the geometric priors, effectively preventing the network from being misled by inherent noise in MDE maps. To our knowledge, such a decoupled fusion and refinement paradigm has yet to be fully explored in RSCD.

The remainder of this article is organized as follows. Section II reviews the related work. Section III details the KGDNet framework and its key components. Section IV presents the benchmark datasets, evaluation metrics, experimental results, and ablation studies. Section V discusses these results. Section VI concludes this article and outlines future work.

II. Related Work

RSCD aims to identify land-cover transitions between bitemporal very high-resolution (VHR) images. Existing studies can be broadly grouped into two paradigms: image-feature-driven RSCD, which learns discriminative representations directly from bitemporal inputs, and prior-knowledge-guided RSCD, which incorporates external cues to stabilize feature learning and suppress pseudo-changes induced by illumination, seasonal, or registration variations. Recent advances in MDE foundation models further enrich this latter paradigm by providing annotation-free geometric priors directly from optical imagery.

A. Image-Feature-Driven RSCD

1) Convolutional Neural Network-Based Methods: Convolutional neural networks (CNNs) were among the earliest deep learning architectures applied to RSCD. Early works mainly adapted fully convolutional networks (FCNs) with encoder–decoder structures to learn bitemporal differences [2], [25], [26]. Daudt et al. [19] proposed FC-EF, FC-Siam-Conc, and FC-Siam-Diff to explore different fusion strategies in Siamese networks, investigating when and how to integrate bitemporal information. However, such designs largely rely on generic feature fusion and may underexploit high-level change semantics.

With increasing image resolution and scene complexity, single-scale convolutions struggle to simultaneously capture small objects and large structures, which motivates the adoption of multiscale representations and enlarged receptive fields. Peng and Kim [27] incorporated multiscale representations into U-Net, while Choi and Kim [28] employed dilated convolutions to capture long-range context. To preserve fine boundaries in VHR images, decoders increasingly emphasize boundary refinement and contextual aggregation. SNUNet [29] densely fuses shallow and deep features with channel attention, while SEIFNet [30] enhances spatiotemporal difference modeling and suppresses pseudo-changes via adaptive context fusion.

Despite these advances, CNNs remain limited in modeling long-range dependencies [31]. Under illumination variations, seasonal changes, or viewpoint shifts, CNN-based models often confuse pseudo-changes with real ones. This limitation becomes more severe as spatial resolution increases, motivating architectures with stronger global modeling capability and better efficiency tradeoffs [23].

2) Transformer-Based Methods: Transformers mitigate CNN locality via self-attention, enabling explicit modeling of global pixel relations. Transformer-based RSCD methods generally fall into three categories: pure Transformers, hierarchical Transformers, and hybrid CNN–Transformer architectures.

a) Pure transformers: A representative work in this category, BIT compresses bitemporal images into a compact set of semantic tokens for attention in a low-dimensional space, enabling token-pixel interactions that retain global context while reducing computation [20]. Complementarily, ChangeFormer by Bandara and Patel [32] adopts a hierarchical Transformer encoder with a lightweight MLP decoder in a Siamese setup, highlighting the value of multiscale long-range modeling for fine-grained boundaries and weak-change regions.

b) Hierarchical transformers: To meet the compute and memory constraints of high-resolution inputs, hierarchical and window-based designs have been explored [33]. SwinSUNet [34] employs a multistage architecture with window self-attention, avoiding CNN locality issues while preserving pyramid representations, which benefits the delineation of regular objects such as buildings. Similarly, Teng et al. [35] designed a foreground-aware module (FAF) to strengthen representations in dense small-object scenarios and better balance global and local cues.

c) Hybrid CNN–transformer designs: Pure-attention models impose heavy computation on VHR imagery, and CNNs capture global context poorly. Therefore, hybrid designs balance accuracy, efficiency, and robustness [36], [37]. Liu et al. [38] propose AMTNet, which exchanges features to reduce interdomain discrepancies and strengthens multiscale representations to detect changes across diverse object sizes. Tan et al. [39] introduce a progressive feature refinement module (PFRM) that targets hard examples, constrains parameter growth, and accelerates inference.

Overall, Transformer-based approaches address CNNs' shortcomings in long-range and cross-scale consistency through global self-attention [40], tokenization, and hierarchical strategies. Window attention and hybridization further improve the efficiency-detail tradeoff. Nevertheless, for VHR images or long temporal sequences, pure attention may still be memory- and compute-intensive, limiting scalability in large deployments [41].

3) Mamba-Based Methods: Both CNNs and Transformers face limitations in modeling long-range dependencies and efficiency. Recently, Mamba, a state-space model (SSM) architecture with linear complexity, has shown strong capability in long-sequence modeling [42]. In RSCD, initial attempts integrate Mamba to enhance global spatiotemporal interaction. For example, CDMamba [43] extends Vision Mamba [44] for unified spatiotemporal modeling, while other approaches combine Mamba with CNNs to balance global context and local detail.

Despite architectural progress, most image-driven methods still rely primarily on optical appearance differences [22].

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In VHR scenes, such monomodal representations are sensitive to misregistration, illumination, and phenological variations, making reliable discrimination between true and pseudo-changes difficult. This motivates the introduction of prior knowledge to provide more stable structural constraints.

B. Prior-Knowledge-Guided RSCD

1) Edge Knowledge-Based Methods: Conventional optical-image pipelines often suffer from blurred or eroded boundaries, particularly in VHR imagery where intricate edges tend to be lost during feature extraction and decoding [31], [45], [46]. Joint modeling of edge detection and change detection (CD) can strengthen boundary awareness while predicting change regions. For instance, edge-aware modules embedded in the encoder and joint supervision at the decoder have been used to enhance boundary completeness in building change tasks, and Li et al. [14] explicitly formulated sample edges as an auxiliary task with multiscale edge supervision to improve contour integrity and the separability of small-object boundaries.

However, edge priors mainly enhance local boundary sharpness and do not fundamentally address pseudo-changes arising from radiometric or phenological variations since they lack explicit geometric or physical constraints.

2) Geometric Knowledge-Based Methods: Geometric priors introduce stable physical constraints beyond 2-D appearance by injecting cues such as relative height, spatial connectivity, and regional consistency into feature learning. These cues help alleviate sensitivity to illumination variations, shadows, seasonal vegetation, and minor misregistration. Representative approaches exploit DEM guidance [47] or impose geometric-consistency losses [48] to suppress pseudo-changes from nontarget entities, generally improving robustness and generalization. More recent works include Xie et al. [49], who incorporated height-difference learning as a dedicated branch, and Liu et al. [50], who fused digital surface model (DSM) and aerial imagery in a cross-modal Transformer to jointly optimize semantic change and height variation. Extending geometric priors to vector data, Chen et al. [13] proposed ObjFormer, which leverages structural priors from OSM to guide token generation in Transformers, further highlighting the value of geometry-aware representations.

However, these traditional approaches are largely constrained by the availability of high-quality external data. Obtaining accurate DSMs or DEMs typically requires expensive acquisition campaigns or dense point clouds, and the high cost of annotation limits scalability in large-area RSCD applications [51], [52]. To alleviate this dependency, recent studies have shifted toward leveraging general-purpose MDE models. These data-driven approaches provide an annotation-free alternative by generating geometric priors directly from single optical images, thereby bypassing the need for curated elevation products.

Despite these advantages, geometric priors are not immune to imperfections. In practical scenarios, interference from vegetation, water surfaces, or shadow effects may contaminate geometric cues [10], [53]. Naively injecting such priors into CD models can therefore introduce additional pseudo-changes, making robust alignment and refinement mechanisms necessary to adaptively mitigate prior-induced noise.

3) Methods Based on Prior Knowledge From Other Domains: Beyond explicit edge and geometric cues, a growing body of literature focuses on knowledge-guided computation for RSCD, utilizing priors from heterogeneous sources to regularize learning. One representative paradigm is transfer learning and knowledge distillation. For instance, Liu et al. [54] proposed a cross-task transfer strategy that leverages structural priors from a building extraction task to facilitate CD in VHR imagery. Similarly, in semi-supervised scenarios, teacher–student networks have been widely adopted to distill reliable knowledge from labeled data to unlabeled samples [55]. To further lower annotation costs, implicit priors via adversarial class prompting have been explored to guide weakly supervised CD [56].

Another significant trend involves leveraging heterogeneous and multimodal priors. To bridge the gap between different sensors or modalities, recent works have exploited structural consistency in the Fourier domain [57] or distilled modality-invariant commonality representations [58] for unsupervised CD. At a higher semantic level, vision-language pretraining has been introduced to RSCD. Methods like ChangeCLIP [59] and CVNet [60] align visual features with linguistic concepts, injecting semantic priors that enhance robustness against pseudo-changes caused by seasonal or illumination variations.

C. MDE Foundation Models

MDE, which aims to predict dense depth maps from single RGB images, is a fundamental computer vision task. As our work leverages MDE to generate external geometric priors, we briefly review recent advances that enable reliable, domain-agnostic geometry extraction.

Early deep learning methods primarily formulated MDE as a supervised regression problem on specific datasets. While pioneering works like MiDaS [61] introduced multi-dataset training to establish a general-purpose baseline, they often struggled with blurred boundaries and the loss of fine structural details in complex scenes. Recently, MDE has entered the foundation model era. Driven by massive training data and scaled capacities, architectures such as Depth Pro [16], VGGT [18], and Depth Anything V2 (DAv2) [17] have achieved unprecedented leaps in geometric consistency, boundary sharpness, and zero-shot generalization. For remote sensing applications, these advanced models can reliably recover relative elevations and spatial layouts from top-down imagery without requiring domain-specific fine-tuning.

Motivated by these advancements, recent RSCD studies have made pioneering explorations into integrating depth priors. For instance, ChangeDA [62] ingeniously incorporates depth as a subtask within a multitask framework, utilizing differential depth augmentation to supervise the network. Similarly, DepthCD [63] employs pretrained MDE models to generate pseudo-labels for semi-supervised distillation. While these methods have demonstrated remarkable effectiveness, their task-level coupling paradigm generally treats the MDE-generated depth as a direct supervision target. This implies

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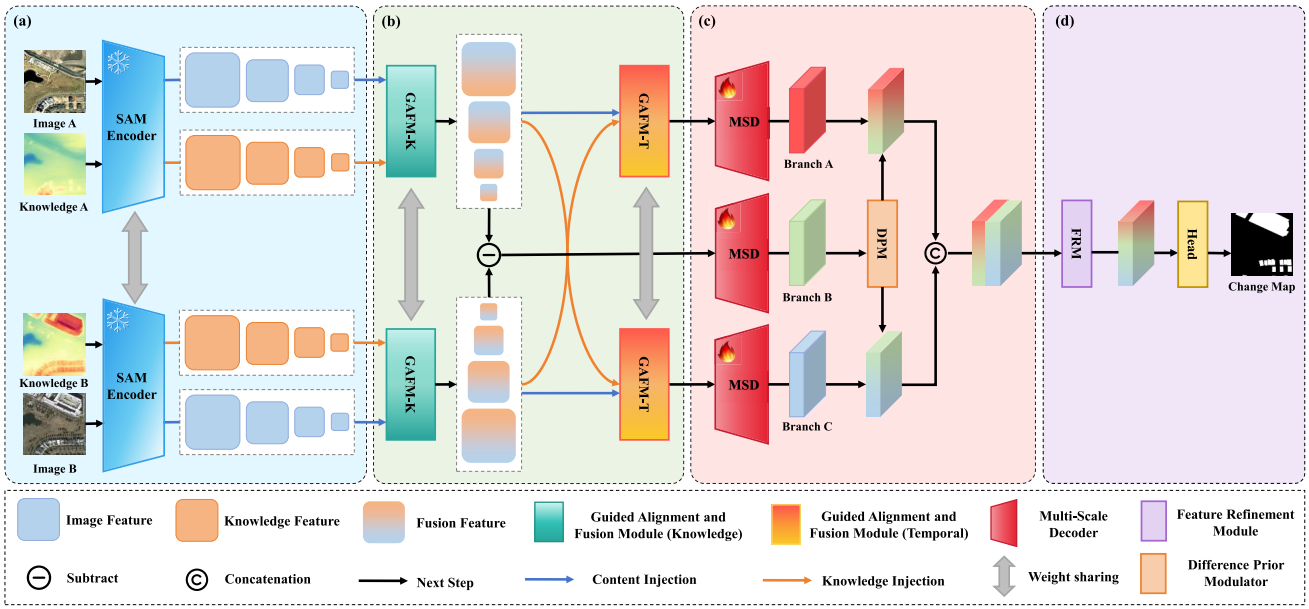


Fig. 2. Overall architecture of KGDNet. The framework comprises four parts. (a) Multimodal feature encoding—data from different modalities are encoded by weight-sharing encoders. (b) Guided cross-dimensional aggregation—centered on GAFM, which aggregates both cross-modal and cross-temporal features. (c) Tri-branch decoupled modeling—change cues are decoupled and fed into separate decoders to learn deep change mechanisms. (d) Coupled feature refinement—mechanism-specific features are integrated and refined to produce the final change map.

a high reliance on the accuracy of the pseudo-depth maps, which may become sensitive to the inherent noise or artifacts occasionally produced by MDE models in complex scenes. Furthermore, employing dual-encoder or ViT-based distillation architectures typically requires considerable computational resources.

Building upon these valuable insights, KGDNet explores another path: a decoupled multimodal fusion paradigm. Rather than treating depth prediction as an intrinsic task target, KGDNet utilizes MDE-derived depth maps as extrinsic knowledge guidance. Specifically, we introduce a dedicated refinement mechanism to dynamically assess and refine the geometric cues, enabling the network to adaptively mitigate potential MDE-induced artifacts. Furthermore, by explicitly decoupling the change mechanisms into a physically interpretable tri-branch architecture, KGDNet is able to employ lightweight components. This design shifts RSCD toward a flexible, knowledge-guided framework, offering an efficient and robust complement to existing depth-integrated architectures.

III. Methodology

This section details the architecture of the proposed KGDNet. We first present the overall framework, followed by in-depth descriptions of its four key stages: 1) multimodal feature encoding; 2) guided cross-dimensional aggregation; 3) tri-branch decoupled modeling; and 4) coupled feature refinement.

A. Overall Architecture

To transcend the limitations of conventional binary CD and effectively mitigate pseudo-changes, we propose KGDNet, an end-to-end network designed for a nuanced analysis of change mechanisms. As depicted in Fig. 2, the architecture is systematically organized into four stages.

During the multimodal feature-encoding stage, KGDNet processes a bitemporal optical-image pair and corresponding single-channel depth maps; the latter are generated offline by the DAv2 [17] model. We adopt a frozen FastSAM [64] backbone as our visual encoder. It is worth noting that while FastSAM inherits the class-agnostic representation capabilities of the segment anything model (SAM) [65], it is fundamentally designed as a lightweight convolutional architecture, which is significantly more compact than models like ViT-B. This strategic choice allows KGDNet to extract transferable and robust feature embeddings [66] without incurring the prohibitive computational overhead associated with massive foundation models, ensuring high efficiency for practical RSCD deployment.

The core of our framework is the guided cross-dimensional aggregation stage, which leverages our novel GAFM. It operates in two distinct modes. GAFM-K (knowledge-guided) injects geometric information by fusing depth-derived features with optical features for each time point, thereby generating 3-D geometry-aware hybrid representations. Subsequently, GAFM-T (temporal-guided) executes cross-temporal interaction between these hybrid representations to achieve precise alignment and accentuate salient differences. On these aligned hybrid features, the MH-LSTA module conducts latent-space attention to capture long-range spatial dependencies and temporal dynamics. Both GAFM-K and GAFM-T employ a Siamese, weight-sharing design to enhance parameter efficiency and promote robust learning of alignment patterns.

The aligned features are subsequently channeled into the tri-branch decoupled modeling stage. Here, the complex CD

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task is explicitly decomposed and routed into three specialized parallel branches to model distinct heterogeneous change patterns. This explicit decoupling facilitates a more targeted and comprehensive feature characterization of each mechanism.

Finally, in the coupled feature refinement stage, outputs from the three specialized branches are systematically integrated by the feature refinement module (FRM). The FRM resolves potential interbranch ambiguities and sharpens predicted boundaries before a lightweight head produces the final binary change map. This holistic design enables the robust discrimination of genuine land-cover evolution within a unified, geometry-aware feature space.

B. Multimodal Feature Encoding

To mitigate pseudo-changes induced by illumination, shadows, or phenology, KGDNet adopts a prior-injection multimodal feature encoding scheme. This stage involves two primary steps.

First, we leverage the DAv2 model to generate single-channel depth maps $\{D_{\mathrm{T}_1}$ and $D_{\mathrm{T}_2}\}$ from the input optical images, serving as label-free geometric priors. To ensure seamless integration with the RGB-native backbone, we adapt these single-channel maps into a three-channel format via pseudo-color mapping. First, the raw depth values D are normalized to a [0, 1] range

$$\tilde{D} = \frac{D - D_{\min}}{D_{\max} - D_{\min} + \epsilon} \tag{1}$$

where $D_{\min}$ and $D_{\max}$ are the minimum and maximum depth values in the map and ϵ is a small constant for numerical stability. A fixed color mapping function $\mathcal{F}(\cdot)$ then projects the normalized map into the RGB space

$$D^{\mathrm{RGB}} = \mathcal{F}\left(\tilde{D}\right), \quad D^{\mathrm{RGB}} \in \mathbb{R}^{H\times W\times 3}. \tag{2}$$

Second, due to the domain discrepancy between optical imagery and depth-derived pseudo-color representations, conventional backbones often yield suboptimal cross-modal alignment. Accordingly, we adopt a lightweight FastSAM encoder, denoted $\mathcal{E}_\theta$, owing to its demonstrated cross-domain generalization. We instantiate $\mathcal{E}_\theta$ in a Siamese, weight-sharing form to process the optical inputs $\{I_{\mathrm{T}_1}, I_{\mathrm{T}_2}\}$ and the corresponding pseudo-color depth inputs $\{D^{\mathrm{RGB}}_{\mathrm{T}_1}, D^{\mathrm{RGB}}_{\mathrm{T}_2}\}$. During training, the encoder parameters are kept fixed to retain generic features and to mitigate overfitting to artifacts in the estimated depth maps. Given these inputs, $\mathcal{E}_\theta$ produces a four-level feature pyramid

$$\begin{aligned}\left\{S^i\right\}_{i=1}^{4} &= \mathcal{E}_\theta\left(I_{\mathrm{T}_1}, I_{\mathrm{T}_2}, D^{\mathrm{RGB}}_{\mathrm{T}_1}, D^{\mathrm{RGB}}_{\mathrm{T}_2}\right)\\ \left\{S^i\right\} &\in \mathbb{R}^{\left(H/2^{i+1}\right)\times\left(W/2^{i+1}\right)\times C_i}\end{aligned} \tag{3}$$

where i indexes the pyramid levels, S_i denotes the feature map at the corresponding scale, the spatial resolution is downsampled by 2^{i+1} relative to the input, and the channel widths are $\{C_i\} = \{160, 320, 640, 640\}$ from fine to coarse.

C. Guided Cross-Dimensional Aggregation

This stage aggregates information across modalities, time, and scales under the guidance of the priors from multimodal feature encoding. GAFM aligns and fuses features to reduce cross-modal and cross-temporal discrepancies, while the MH-LSTA performs efficient latent-space attention to capture long-range spatiotemporal context. The result is a registration-robust, context-rich representation passed to the tri-branch module.

1) Guided Alignment and Fusion Module: The central challenge in knowledge-guided CD lies in integrating heterogeneous information sources—namely, cross-modal geometric priors and cross-temporal appearance features—within a single, efficient mechanism. To this end, we introduce GAFM, a unified architecture designed for knowledge-guided feature alignment and temporal interaction. The overall structure of GAFM is illustrated in Fig. 3.

GAFM operates on a pair of inputs, where F_{A} denotes the content feature to be preserved and F_{B} denotes the feature providing auxiliary guidance. The computational process is formulated as

$$F_{\mathrm{con}} = \mathrm{SpDown}\left(F_{\mathrm{A}}\right) \tag{4}$$

$$F_{\mathrm{guide}} = \mathrm{SpDown}\left(F_{\mathrm{B}}\right) \tag{5}$$

$$F_{\mathrm{LSTA}} = \text{MH-LSTA}\left(F_{\mathrm{con}}, F_{\mathrm{guide}}\right) \tag{6}$$

$$\tilde{F}_{\mathrm{LSTA}} = \mathrm{SpUp}\left(F_{\mathrm{LSTA}}\right) \tag{7}$$

$$F_{\mathrm{out}} = \mathrm{LN}\left(F_{\mathrm{A}} + \mathrm{Gate}\left(\tilde{F}_{\mathrm{LSTA}}\right) \odot \tilde{F}_{\mathrm{LSTA}}\right) \tag{8}$$

where SpDown denotes a 3×3 convolution followed by batch normalization and a GELU activation and SpUp denotes bilinear upsampling to restore the original spatial resolution. The Gate$(\cdot)$ module is a channel-wise gating branch composed of adaptive average pooling, two 1×1 convolution layers with an intermediate GELU activation, and a final sigmoid function. Layer normalization LN$(\cdot)$ is applied along the channel dimension. Through the gated residual connection, the modulated interaction features are adaptively fused into the content stream.

This unified formulation can be instantiated as GAFM-K for cross-modal alignment or GAFM-T for cross-temporal fusion by appropriately assigning F_{A} and F_{B}.

At a structural level, GAFM shares a coarse resemblance with existing feature fusion modules that follow a downsampling–interaction–upsampling paradigm with channel-wise gating. However, it is not a direct instantiation of these designs. Instead, GAFM is tailored for knowledge-guided RSCD and differs from conventional attention- or gating-based modules in both design motivation and functional role.

Specifically, classical channel-attention modules such as squeeze-and-excitation (SE) [67] blocks and convolutional block attention module (CBAM) [68] adopt a self-modulation paradigm, where channel-wise importance is inferred solely from the input feature itself. As a result, feature recalibration is performed independently within a single stream. In contrast, GAFM introduces an explicit guidance pathway, in which auxiliary features—derived from geometric priors or cross-temporal references—are leveraged to modulate the content stream through asymmetric cross-feature interaction.

Similarly, nonlocal blocks and Transformer-style self-attention model pairwise relations among homogeneous

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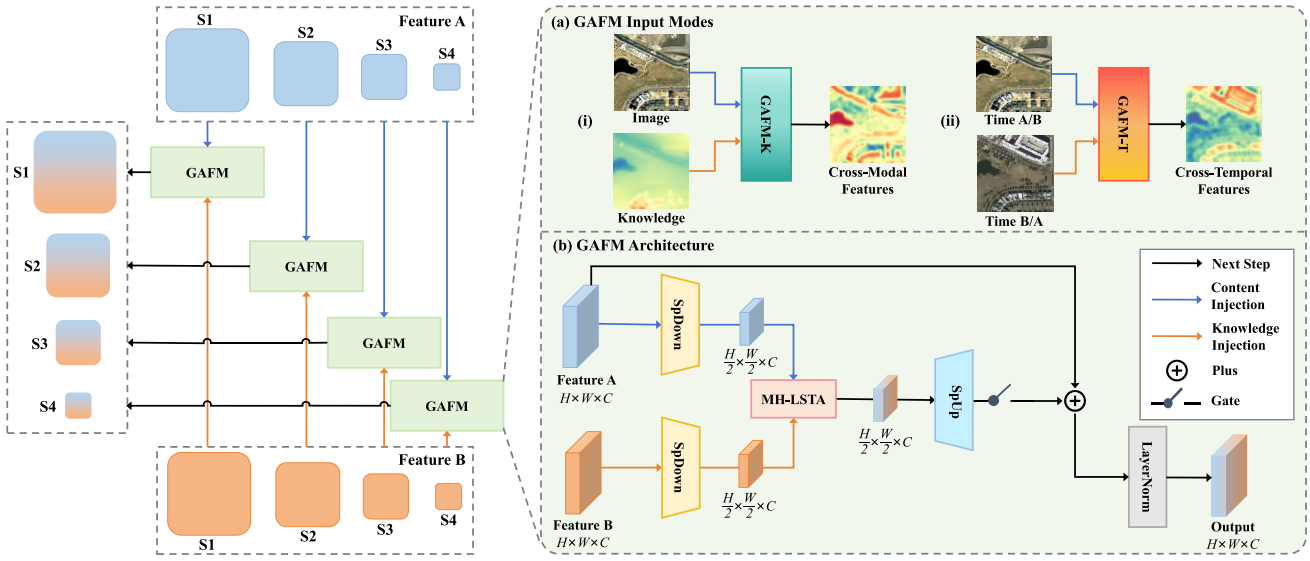


Fig. 3. Architecture of the GAFM. The diagram at left illustrates the application of GAFM across a multiscale feature pyramid. The diagram at right details two primary aspects. (a) Two distinct input modes, which include GAFM-K for cross-modal alignment between image features and knowledge priors and GAFM-T for cross-temporal feature alignment. (b) Internal component architecture, showing that the data flow through spatial downsampling (SpDown), the MH-LSTA attention block, spatial upsampling (SpUp), and a gated residual connection for feature fusion.

tokens, treating all inputs as semantically equivalent. GAFM departs from this paradigm by adopting a content-guidance formulation, where MH-LSTA explicitly models cross-source dependencies rather than self-relations, which is more suitable for alignment between heterogeneous feature sources.

Moreover, GAFM decouples global interaction modeling and feature fusion into two stages: cross-source dependencies are first captured in a latent space by MH-LSTA, and the final feature integration is subsequently regulated by a gated residual pathway. This design separates relation learning from feature injection, allowing flexible control over guided modulation.

Finally, GAFM adopts a unified formulation for both cross-modal and cross-temporal alignment. Through different input assignments, GAFM-K and GAFM-T share the same architecture, enabling consistent alignment mechanisms across modalities and time, which is rarely explored in existing attention or channel-gating modules.

2) Multihead Latent Spatiotemporal Attention: The quadratic complexity of standard self-attention with respect to token count poses a significant computational barrier for high-resolution remote sensing imagery. To address this challenge, we propose the MH-LSTA, a guidance-aware latent-space attention module that preserves global context modeling while remaining computationally feasible for high-resolution remote sensing imagery. The module combines two complementary ideas: latent space compression and a lightweight, parameter-free Geo-PE, as illustrated in Fig. 4.

Attention is first computed in a compressed latent space by projecting the key (**K**) and value (**V**) tensors—derived from the content feature (F_{A})—with a 1×1 convolution that reduces the channel width by a factor r

$$\{\mathbf{K},\mathbf{V}\}=\mathrm{Conv}_{1\times1}(F_{\mathrm{A}})\in\mathbb{R}^{H\times W\times(C/r)}. \tag{9}$$

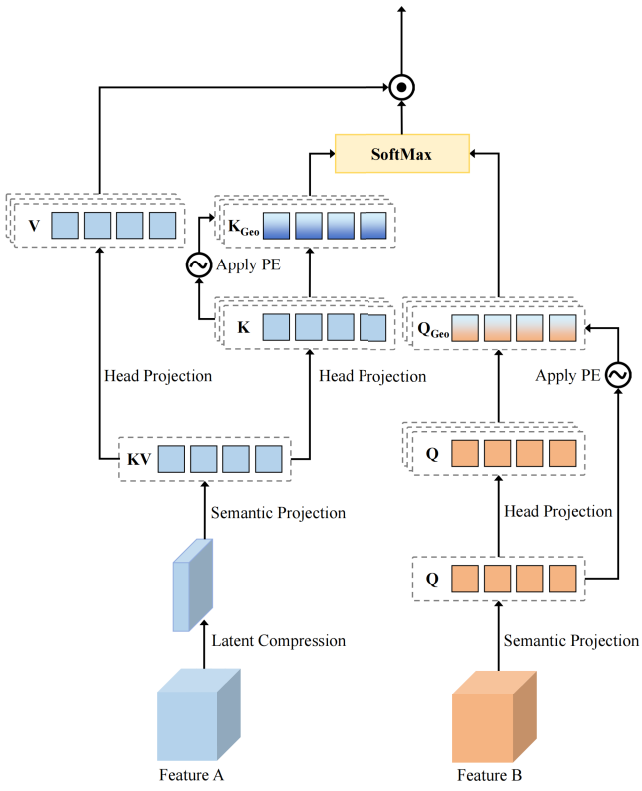


Fig. 4. Architecture of the MH-LSTA. The key and value matrices, **K** and **V**, are projected from F_A by a convolution to C/r channels, whereas the query matrix **Q** is computed from F_B at full size. A lightweight Geo-PE modulates **Q** and **K** before the scaled dot-product attention. The attended output is added residually to F_B to form the module output, enabling global context modeling with reduced computation.

In contrast, the query (**Q**) tensor—derived from the knowledge feature F_{B}—is kept at full width. This asymmetric design preserves the granularity of the query pathway while

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substantially reducing computation and memory in the key–value pathway.

We generate Geo-PE on demand to accommodate arbitrary spatial resolutions. Let $u \in \{0, \ldots, \text{H-1}\}$ and $v \in \{0, \ldots, \text{W-1}\}$ denote grid coordinates and d denote the channel width of the positional encoding. We set $d = C/r$ so that it matches the compressed channel width used for the key–value pathway. The frequency bank and the 1-D sinusoidal mapping are defined as

$$\omega = \left[\, 10000^{-k/d} \,\right]_{k=0}^{\lceil d/2 \rceil - 1} \tag{10}$$

$$\Phi(t) = [\sin(t\,\omega) \quad \cos(t\,\omega)] \in \mathbb{R}^d \tag{11}$$

which defines the notation $\Phi(\cdot)$ and provides position-sensitive modulation with negligible overhead. The 2-D encoding is the sum of the horizontal and vertical components

$$PE_{\text{Geo}}(u, v) = \Phi(u) + \Phi(v) \in \mathbb{R}^d. \tag{12}$$

We then modulate $\mathbf{Q}$ and $\mathbf{K}$ by channel-wise scaling, and the scaling factors are broadcast over the spatial dimensions

$$\mathbf{K}_{\text{Geo}} = \text{Concat}\left[\mathbf{K}, \mathbf{K} \odot (1 + PE_{\text{Geo}}(u, v))\right] \tag{13}$$

$$\mathbf{Q}_{\text{Geo}} = \text{Concat}\left[\mathbf{Q}, \mathbf{Q} \odot (1 + PE_{\text{Geo}}(u, v))\right] \tag{14}$$

where $\odot$ denotes the element-wise multiplication and Concat$[\cdot]$ concatenates along the channel dimension.

Finally, the output is obtained by scaled dot-product attention

$$F_{\text{out}} = \text{Softmax}\left(\frac{\mathbf{Q}_{\text{Geo}}\mathbf{K}_{\text{Geo}}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{15}$$

where d_k is the channel width of $\mathbf{K}_{\text{Geo}}$. This design enhances sensitivity to absolute and relative positions with minimal computational cost while retaining the benefits of global context modeling under a feasible computational budget.

D. Tri-Branch Decoupled Modeling

Conventional CD frameworks rely on a monolithic binary paradigm, forcing a single model to learn the heterogeneous patterns underlying different types of change. KGDNet departs from this by explicitly decoupling the task into three specialized branches, each designed to capture a fundamental change mechanism.

1) Multiscale Decoder: Each branch adopts a multiscale decoder (MSD) composed of three decoder fusion blocks (DFBs); the architecture is shown in Fig. 5. While progressively upsampling the features, the MSD fuses them with the corresponding cross-temporal features at the same level, so as to preserve high-level semantics and low-level spatial details. The procedure at level i is

$$S_{\text{up}}^i = \text{SpUp}\left(S_{\text{A}}^i\right), \qquad i \in \{2, 3, 4\} \tag{16}$$

$$S_{\text{cat}}^i = \text{Concat}\left[S_{\text{A}}^{i-1}, S_{\text{up}}^i\right] \tag{17}$$

$$S_{\text{DFB}}^i = \text{Fusion}\left(S_{\text{cat}}^i\right) + \text{Adaptor}\left(S_{\text{B}}^{i-1}\right) \tag{18}$$

where Fusion consists of two basic 3×3 depthwise separable convolution blocks, each block containing a convolution layer, a batch normalization layer, and a GELU activation; Adaptor is a basic 1×1 convolutional block with the same composition; and SpUp upsamples by a factor of 2 using bilinear interpolation.

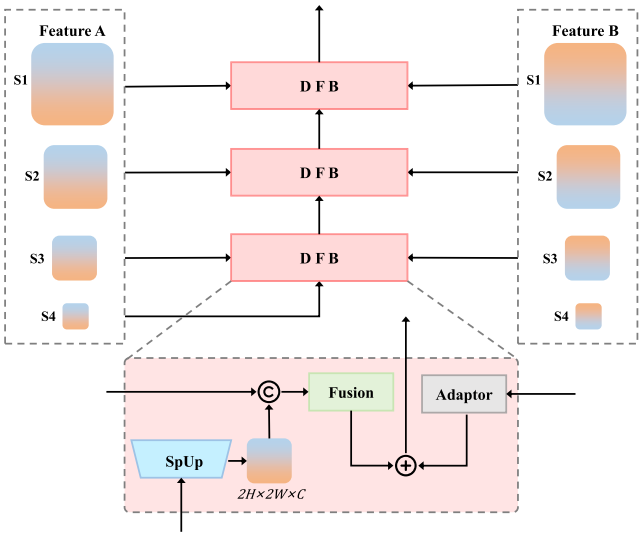


Fig. 5. Architecture of the MSD. At four scales (S1–S4), upsampled features are fused with same-scale features via a DFB, while an adaptor injects cross-temporal cues from the difference branch. This coarse-to-fine fusion preserves high-level semantics and sharp boundaries, yielding refined features for the final change map.

2) Appearance Branch (Branch A): This branch is dedicated to identifying newly emerging objects. It is configured to use the pre-change feature (F_{T_1}) as knowledge (F_{B}) and the post-change (F_{T_2}) as the primary content feature (F_{A}). This directional comparison effectively isolates and highlights features present at T_2 but absent at T_1. GAFM-T output features (F_{A}^i) are decoded by the MSD and then dynamically refined by a confidence map

$$F_{\text{A}}^i = \text{GAFM-T}\left(F_{\text{A}} = F_{\text{T}_2}, F_{\text{B}} = F_{\text{T}_1}\right) \tag{19}$$

$$\tilde{F}_{\text{A}} = \text{MSD}\left(F_{\text{A}}^1, F_{\text{A}}^2, F_{\text{A}}^3, F_{\text{A}}^4\right) \odot M_{\text{con}}. \tag{20}$$

3) Difference Branch (Branch B): This branch serves a dual purpose. It captures explicit structural differences between the bitemporal features, and more crucially, it acts as an active dynamic refinement mechanism for the geometric priors. By introducing the difference prior modulator (DPM), this branch is elevated from a conventional difference calculator to a proactive noise filter within our decoupled architecture.

First, it extracts cross-temporal variation representations by operating on the absolute difference between the bitemporal, geometry-aware hybrid features.

Second, this branch acts as the engine for dynamic prior refinement. Because its input features have already been processed by the GAFM, they inherently contain valuable 3-D-aware cues. However, to prevent potential MDE-induced artifacts from affecting the final decision, the decoded difference feature ($\tilde{F}_B$) is not directly concatenated for the final prediction. Instead, it serves as the exclusive input to the DPM to generate a spatial confidence map. This process is defined as follows:

$$F_{\text{B}}^i = |F_{\text{T}_1} - F_{\text{T}_2}| \tag{21}$$

$$\tilde{F}_{\text{B}} = \text{MSD}\left(F_{\text{B}}^1, F_{\text{B}}^2, F_{\text{B}}^3, F_{\text{B}}^4\right) \tag{22}$$

$$M_{\text{con}} = \text{DPM}\left(\tilde{F}_{\text{B}}\right) \tag{23}$$

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where DPM(·) is implemented as a lightweight convolutional head. It processes the multiscale difference feature $\tilde{F}_{\mathrm{B}}$ through a 3×3 convolution with GELU activation, followed by a 1×1 convolution that projects the features into a single-channel map. A final sigmoid activation function normalizes this map to the range [0, 1] to form the spatial confidence map M_{con}.

Crucially, this confidence map acts as a spatial gating mechanism. It allows the network to adjudicate ambiguities and explicitly suppress spurious geometric responses—such as noisy depth cues from nontarget elements like trees or shadows—that may have persisted in the other branches.

4) Disappearance Branch (Branch C): Symmetrical to the appearance branch, this branch focuses on objects that have been removed or have vanished. It uses the T_1 feature as the primary content, guided by the T_2 feature, thereby isolating signatures that existed only at T_1. Similar to Branch A, its output is also modulated by the DPM-generated confidence map

$$F_{\mathrm{C}}^{i} = \text{GAFM-T}\left(F_{\mathrm{A}} = F_{\mathrm{T}_1}, F_{\mathrm{B}} = F_{\mathrm{T}_2}\right) \tag{24}$$

$$\tilde{F}_{\mathrm{C}} = \text{MSD}\left(F_{\mathrm{C}}^{1},\ F_{\mathrm{C}}^{2},\ F_{\mathrm{C}}^{3},\ F_{\mathrm{C}}^{4}\right) \odot M_{\mathrm{con}}. \tag{25}$$

E. Coupled Feature Refinement

The final stage integrates the outputs from the three decoupled branches to generate the final change prediction via fusion, refinement, and prediction.

First, the confidence-gated features from the appearance branch ($\tilde{F}_{\mathrm{A}}$) and disappearance branch ($\tilde{F}_{\mathrm{C}}$) are concatenated along the channel dimension to form a unified representation. Then, the fused features are refined by an FRM composed of three residual refinement blocks (RRBs), which models cross-branch interactions and enhances change boundaries. Finally, a lightweight prediction head produces a single-channel probability map that is upsampled to the original resolution.

The process is formulated as

$$F_{\mathrm{cat}} = \text{Concat}\left[\tilde{F}_{\mathrm{A}},\ \tilde{F}_{\mathrm{C}}\right] \tag{26}$$

$$F_{\mathrm{ref}} = \text{FRM}\left(F_{\mathrm{cat}}\right) \tag{27}$$

$$\mathbf{Y} = \text{Upsample}\left(\sigma\left(\text{Head}\left(F_{\mathrm{ref}}\right)\right)\right) \tag{28}$$

where Concat[·] denotes channel-wise concatenation that aggregates the two gated perspectives. FRM is an FRM composed of three RRBs, which learn cross-branch dependencies and resolve inconsistencies to sharpen object boundaries. Head(·) is a lightweight prediction head consisting of two 3×3 convolutions followed by a final 1×1 convolution that outputs a logit map. $\sigma(\cdot)$ denotes the sigmoid function that converts logits to probabilities, and Upsample(·) restores the probability map to the input spatial resolution by bilinear interpolation. The resulting $\mathbf{Y} \in [0,1]^{H \times W}$ serves as the final change-probability map.

IV. Experiments and Results

In this section, we conduct a series of comprehensive experiments to rigorously evaluate the performance of our proposed KGDNet. We first introduce the benchmark datasets, evaluation metrics, and implementation details. Subsequently, we compare KGDNet against representative CD algorithms, presenting both quantitative and qualitative results to demonstrate its effectiveness and robustness. Finally, we perform extensive ablation studies to verify the contribution of each key component.

A. Datasets

To ensure a fair and comprehensive comparison, we evaluate our model on four widely used, high-resolution public benchmark datasets covering both building CD and diverse urban land-cover changes.

1) *LEVIR-CD [69]:* This dataset is a large-scale building CD benchmark containing 637 pairs of VHR Google Earth images with 0.5-m spatial resolution. It spans diverse scenes, building types, and seasonal conditions, capturing various change patterns such as new construction and demolition. The official split provides 445 training, 64 validation, and 128 test pairs, and each image is 1024×1024 pixels. To accommodate GPU memory limitations, we train on random 256×256 crops extracted from the original images.
2) *WHU-CD [70]:* This dataset comprises a single pair of aerial images with 0.2-m resolution and dimensions of $32\,507 \times 15\,354$ pixels, covering Christchurch, New Zealand, and focusing on building changes after the earthquake. The dataset is provided as nonoverlapping 256×256 patches, split into 6096 training, 762 validation, and 762 test pairs.
3) *LEVIR+-CD:* This dataset is an extended version of [69], built from VHR Google Earth images with a spatial resolution of 0.5 m. It contains more complex scenes and subtle building changes than LEVIR-CD. The dataset includes 985 bitemporal image pairs with a size of 1024×1024 pixels and is officially divided into 637 training pairs and 348 test pairs. During training, the images are cropped into 256×256 patches to fit GPU memory constraints.
4) *SYSU-CD [71]:* It is a large-scale urban CD dataset collected from high-resolution aerial images with a spatial resolution of about 0.5 m. Unlike the above datasets that mainly focus on building changes, SYSU-CD covers more diverse change categories in complex urban scenes, including buildings, roads, vegetation, and other land-cover changes. The dataset contains 20 000 bitemporal image patches of size 256×256 pixels, which are split into 12 000 training, 4000 validation, and 4000 test pairs.

B. Evaluation Metrics

To quantitatively evaluate our method, we adopt several standard metrics: overall accuracy (OA), precision (Pre), recall (Rec), $F1$-score (F1), and intersection over union (IoU). For the binary CD task, F1 and IoU are typically the most informative. We also report the mean F1 (mF1) and mean IoU (mIoU), which average the metrics over the change and no-change classes. The metrics are defined as follows:

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{29}$$

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$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{30}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{31}$$

$$\text{F1} = \frac{2 \times \text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{32}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{33}$$

$$\text{mF1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2 \times \text{Pre}_c \times \text{Rec}_c}{\text{Pre}_c + \text{Rec}_c} \tag{34}$$

$$\text{mIoU} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{35}$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. For the class-averaged metrics, $\mathcal{C}$ is the set of classes, and TP_c, FP_c, and FN_c are computed by treating class c as the positive class in the confusion matrix. We report all metrics in percentage; higher values indicate better performance.

C. Implementation Details

All experiments were conducted on an NVIDIA A100 GPU using the PyTorch framework. The depth map priors were generated offline using the DAv2-L model. We employed the binary cross-entropy loss function for supervision. Depending on the specific dataset, we utilized either the SGD optimizer with an initial learning rate of 0.1 or the Adam optimizer with an initial learning rate of $1e^{-4}$. A weight decay of $1e^{-4}$ was applied across all experiments, and the learning rate was adjusted using a cosine annealing schedule. Models were trained for 200 epochs with a batch size of 32. Standard data augmentation techniques, including random flipping, cropping, and rotation, were applied to enhance robustness. All SOTA methods were reproduced based on their official implementations and trained under identical conditions for a fair comparison. Test-time augmentation (TTA) was used during inference to produce more stable results.

D. Comparison With State-of-the-Art Methods

To comprehensively evaluate KGDNet, we selected representative SOTA methods spanning four architectural paradigms.

1) *CNN-Based:* FC-Siam-Conc and FC-Siam-Diff [19] are early Siamese CNN-based methods that explore feature concatenation and explicit difference modeling for CD. STANet [69] introduces spatial–temporal attention within a CNN framework to enhance change representation. SNUNet [29] utilizes nested U-Net and dense connections to enhance multiscale feature fusion. SEIFNet [30] employs dual-branch spatiotemporal difference enhancement and adaptive context fusion to suppress pseudo-changes.
2) *Transformer-Based:* BIT [20] first introduced a Transformer-based design by compressing features into semantic tokens. ChangeFormer [32] uses a pure-Transformer Siamese structure. AMTNet [38] is a hybrid architecture combining CNN local perception with Transformer-based global modeling.
3) *Mamba-Based*: CDMamba [43] merges state-space modeling with convolution.
4) *Prior-Driven*: ChangeDA [62] introduces depth estimation as an auxiliary task to learn implicit 3-D representations, effectively guiding spatial CD. SFEARNet [14] explicitly introduces edge priors as an auxiliary task. CVNet [60] represents a typical foundation model-based CD algorithm, which leverages semantic priors derived from vision-language foundation models to guide change representation learning.

1) Quantitative Experiments: The quantitative comparisons on four high-resolution benchmarks—LEVIR-CD, WHU-CD, LEVIR+-CD, and SYSU-CD—are summarized in Tables I and II. KGDNet consistently achieves strong performance across these diverse datasets, indicating its stable generalization capability in handling scenes ranging from regular building changes to complex urban land-cover evolution.

As presented in Table I, KGDNet achieves the best results on the standard building CD benchmarks, LEVIR-CD and WHU-CD. On LEVIR-CD, it achieves the highest F1 and IoU scores among all compared methods. This advantage can be attributed to the fact that KGDNet decouples single CD into multibranch change modeling, enabling the network to capture fine-grained change representations from complementary perspectives, including appearance, disappearance, and difference patterns. Meanwhile, by incorporating geometry-guided representations, the model perception is extended from conventional 2-D image cues to a more structured 3-D and multiview-aware understanding of urban scenes, which facilitates more accurate localization and delineation of building changes.

On WHU-CD, KGDNet exceeds the second-best method by 0.59% in F1 and 1.06% in IoU. The robustness on this dataset benefits from the geometry-aware feature learning strategy, which introduces structural priors to enhance spatial consistency and preserve object boundaries under viewpoint and scale variations. Such geometry-guided representations allow the network to maintain stable structural perception when observing urban objects from different viewing angles, thereby improving the reliability of change recognition in complex aerial scenarios.

The effectiveness of KGDNet is more evident on the challenging LEVIR+-CD and SYSU-CD datasets, which feature complex urban layouts and diverse change categories, as detailed in Table II. On LEVIR+-CD, KGDNet attains F1 of 85.10% and IoU of 74.06%, exceeding the second-best F1 by 0.99%. This improvement indicates that the integration of geometric priors with decoupled change modeling helps the network better distinguish true structural changes from background variations by leveraging both spatial geometry and semantic appearance cues.

Furthermore, KGDNet also demonstrates highly competitive performance on SYSU-CD. In this scenario, which involves diverse land-cover categories, the tri-branch architecture allows heterogeneous change cues to be modeled separately and then jointly optimized, while the difference branch serves as a dynamic modulation signal to guide feature

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TABLE I
QUANTITATIVE RESULTS ON THE LEVIR-CD AND WHU-CD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Year	LEVIR-CD (%)							WHU-CD (%)						
		OA	Pre	Rec	F1	IoU	mF1	mIoU	OA	Pre	Rec	F1	IoU	mF1	mIoU
FC-Siam-Conc [19]	2018	98.46	88.40	80.22	84.11	72.58	91.65	85.48	97.68	68.07	88.49	76.95	62.53	87.86	80.06
FC-Siam-Diff [19]	2018	98.01	88.51	69.94	78.14	64.12	88.55	81.03	97.58	67.01	88.08	76.11	61.43	87.41	79.45
STANet [69]	2020	98.57	82.50	**91.19**	86.62	76.41	92.94	87.46	98.10	76.02	90.78	82.76	71.87	91.28	84.44
SNUNet [29]	2021	98.99	90.80	89.23	90.01	81.83	94.74	90.39	98.87	85.18	89.72	87.39	77.61	93.40	88.21
BIT [20]	2021	98.92	90.39	89.71	90.05	81.90	94.76	90.42	98.75	86.64	81.48	83.98	72.39	91.11	85.03
ChangeFormer [32]	2022	99.03	91.80	88.94	90.35	82.40	94.92	90.69	99.21	93.71	87.81	90.67	82.93	95.13	91.05
AMTNet [38]	2023	99.05	91.82	89.71	90.76	83.08	94.96	90.76	99.22	92.86	91.99	92.27	85.64	95.48	91.65
SEIFNet [30]	2024	99.09	92.49	89.46	90.95	83.40	95.27	91.11	98.90	83.24	89.85	86.42	76.08	92.88	87.40
ChangeDA [62]	2025	99.22	**93.67**	90.92	92.27	85.65	-	-	**99.54**	**96.45**	91.91	94.12	88.90	-	-
CDMamba [43]	2025	99.06	91.43	90.08	90.75	83.07	95.13	91.04	99.51	95.58	92.01	93.76	88.26	96.19	92.87
SFEARNet [14]	2025	99.07	91.43	90.27	90.85	83.23	95.11	91.01	99.38	94.25	90.55	92.36	85.81	96.37	93.19
CVNet [60]	2025	99.12	92.46	91.07	91.76	84.78	95.41	91.52	99.50	96.33	92.01	94.12	88.89	96.93	94.18
KGDNet (Ours)		**99.23**	93.51	91.12	**92.30**	**85.71**	**95.95**	**92.45**	**99.54**	96.02	**93.44**	**94.71**	**89.96**	**97.24**	**94.74**

TABLE II
QUANTITATIVE RESULTS ON THE LEVIR+-CD AND SYSU-CD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Year	LEVIR+-CD (%)							SYSU-CD (%)						
		OA	Pre	Rec	F1	IoU	mF1	mIoU	OA	Pre	Rec	F1	IoU	mF1	mIoU
FC-Siam-Conc [19]	2018	97.66	71.71	70.21	70.95	54.95	84.87	76.28	89.31	78.91	74.62	76.70	62.21	84.88	74.62
FC-Siam-Diff [19]	2018	97.86	79.51	63.92	70.87	54.88	84.88	76.34	86.40	83.92	52.35	64.48	47.58	78.03	66.03
STANet [69]	2020	98.35	78.83	81.32	80.05	66.74	89.60	82.52	89.70	76.92	80.48	78.66	64.82	85.94	76.06
SNUNet [29]	2021	98.46	81.24	80.75	81.00	68.06	90.10	83.23	88.30	77.38	71.21	74.17	58.94	83.30	72.44
BIT [20]	2021	98.16	80.46	72.46	76.25	61.61	87.65	79.86	89.49	79.97	73.93	76.83	62.38	85.02	74.82
ChangeFormer [32]	2022	98.65	84.21	82.21	83.20	71.23	91.25	84.92	90.16	84.09	71.85	77.49	63.25	85.60	75.70
AMTNet [38]	2023	98.38	78.19	83.41	80.72	67.67	89.93	82.99	89.53	79.29	75.25	77.22	62.89	85.21	75.08
SEIFNet [30]	2024	98.53	83.13	80.21	81.65	68.98	90.44	83.73	91.90	84.81	79.98	82.32	69.96	88.58	79.92
ChangeDA [62]	2025	-	-	-	-	-	-	-	**92.16**	**86.01**	79.72	82.74	70.56	-	-
CDMamba [43]	2025	98.65	85.11	81.00	83.01	70.95	90.89	84.57	88.37	77.87	70.81	74.17	58.95	83.34	72.49
SFEARNet [14]	2025	98.68	82.75	**85.52**	84.11	72.58	91.71	85.61	91.47	84.02	79.94	81.92	69.42	88.14	79.62
CVNet [60]	2025	98.70	85.25	81.90	83.52	71.90	91.35	84.87	90.91	80.67	80.80	80.74	67.70	87.39	78.23
KGDNet (Ours)		**98.79**	**85.39**	84.81	**85.10**	**74.06**	**92.23**	**86.41**	91.92	82.21	**83.91**	**83.05**	**71.01**	**88.87**	**80.47**

interaction between appearance and disappearance branches. Combined with geometry-aware representations, the model forms a multiangle perception of urban structures rather than relying solely on planar image evidence, which helps reduce false detections caused by urban shadows and seasonal vegetation. These results indicate that the proposed geometry-guided and decoupled change modeling framework is not limited to building-only CD but can also be effectively applied to change scenarios involving diverse land-cover categories.

2) Qualitative Experiments: To provide an intuitive and comprehensive comparison, qualitative results of different methods on the test sets of LEVIR-CD, WHU-CD, LEVIR+-CD, and SYSU-CD are presented in Figs. 6–9. The visual comparisons emphasize three key aspects: suppression of pseudo-changes, preservation of structural integrity, and robustness to diverse scene conditions.

As shown in Fig. 6 for the LEVIR-CD dataset, spectral ambiguity poses a significant challenge. In Rows 1, 2, and 4, where building spectral features are highly similar to the ground surface, most competing methods fail to distinguish the targets, resulting in significant missed detections. KGDNet, however, leverages geometric priors to accurately recover these regions. Row 3 highlights the model's boundary precision: while other methods struggle with blurred edges and mistake adjacent ground for changes, KGDNet delineates crisp contours. Notably, Row 5 demonstrates the efficacy of our decoupled modeling: while unified models fail to identify the disappearance of buildings, our dedicated disappearance branch successfully captures this change. Additionally, in Row 6, KGDNet exhibits superior sensitivity to small targets that are overlooked by other approaches.

On the WHU-CD dataset (Fig. 7), two primary issues arise: false alarms due to spectral confusion and fragmentation due to intraobject heterogeneity. As seen in Rows 1, 2, and 5, competing methods frequently misclassify hardened ground as changes due to spectral similarity, whereas KGDNet effectively suppresses these pseudo-changes. Conversely, in

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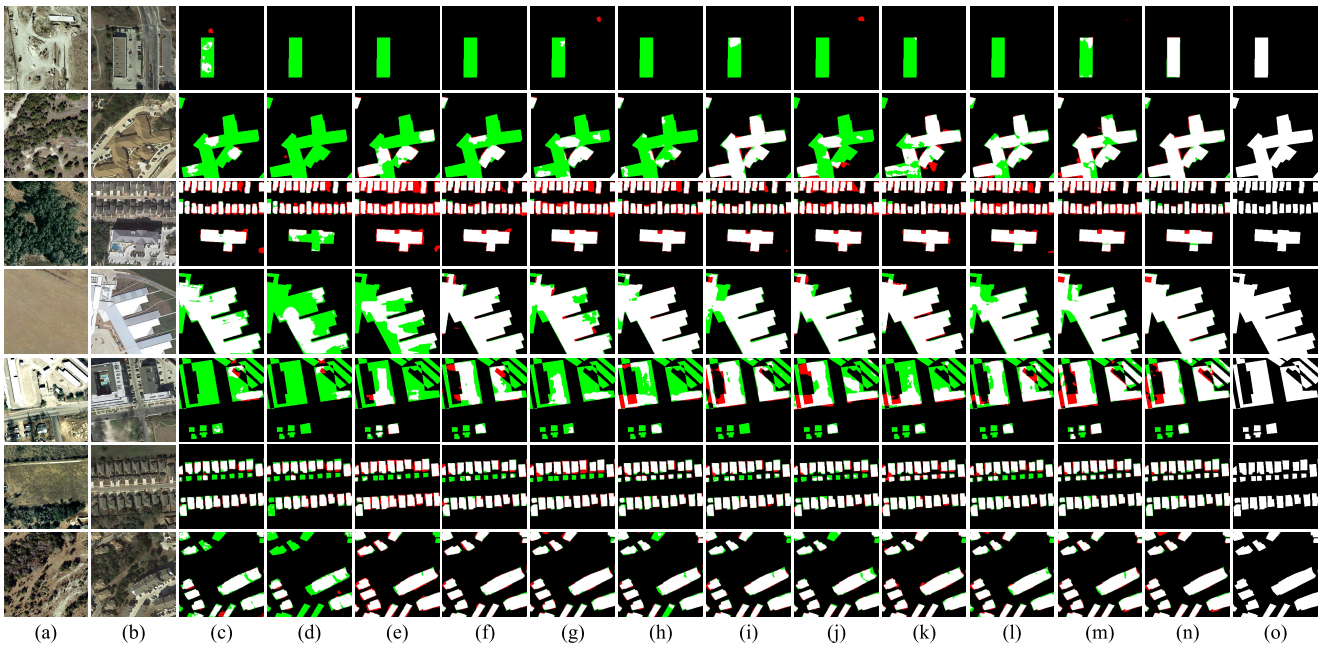


Fig. 6. Visualization results of different methods on the LEVIR-CD test set. (a) T1 image. (b) T2 image. (c) FC-Siam-Conc. (d) FC-Siam-Diff. (e) STANet. (f) SNUNet. (g) BIT. (h) ChangeFormer. (i) AMTNet. (j) SEIFNet. (k) CDMamba. (l) SFEARNet. (m) CVNet. (n) KGDNet. (o) Ground truth. White denotes true positives, black represents true negatives, red indicates false positives, and green indicates false negatives.

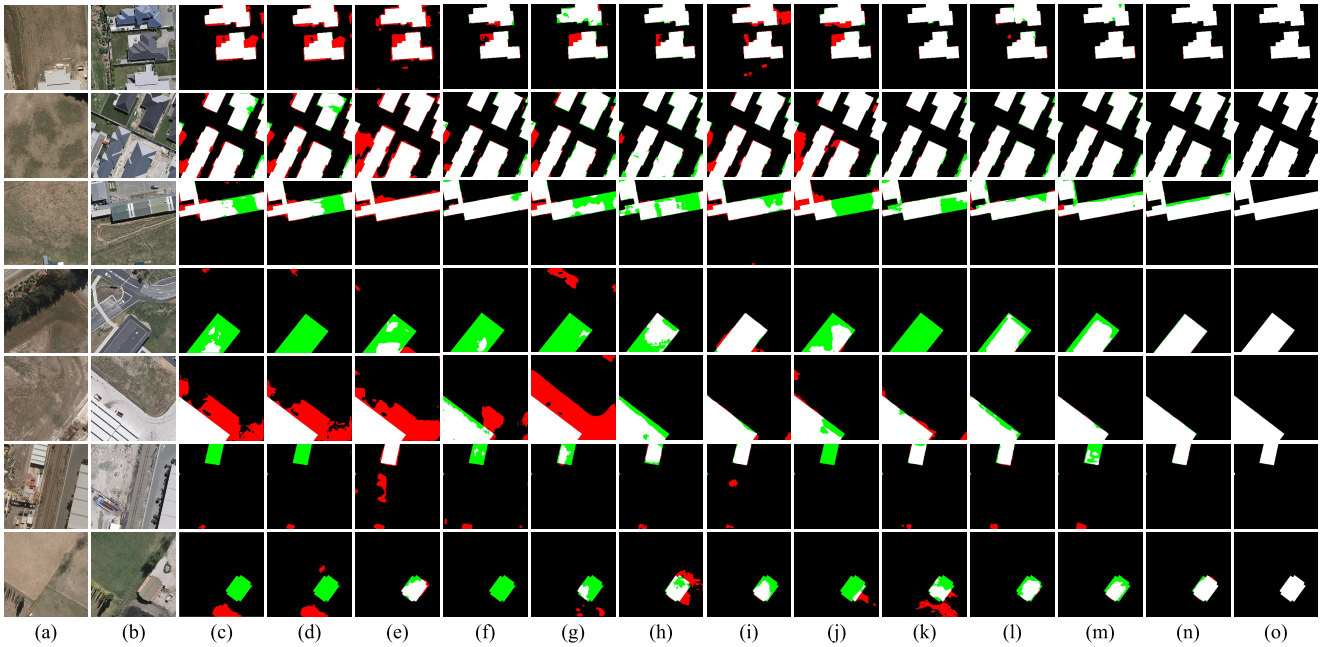


Fig. 7. Visualization results of different methods on the WHU-CD test set. (a) T1 image. (b) T2 image. (c) FC-Siam-Conc. (d) FC-Siam-Diff. (e) STANet. (f) SNUNet. (g) BIT. (h) ChangeFormer. (i) AMTNet. (j) SEIFNet. (k) CDMamba. (l) SFEARNet. (m) CVNet. (n) KGDNet. (o) Ground truth. White denotes true positives, black represents true negatives, red indicates false positives, and green indicates false negatives.

Rows 4 and 6, the similarity between buildings and the background leads to missed detections in baseline models, which KGDNet corrects via geometry-aware feature learning. Furthermore, Row 3 presents a challenge with multicolored roofs. While other methods produce fragmented predictions due to color variations within a single building, KGDNet maintains the structural integrity of the building block, demonstrating robustness against intraclass spectral variance.

For the more challenging LEVIR+-CD dataset (Fig. 8), scene complexity leads to both missed detections and pseudo-changes. In Rows 4 and 7, distinct white buildings with unique structures are often ignored by methods relying solely on 2-D appearance. KGDNet effectively identifies these special targets. Meanwhile, in Rows 1 and 5, complex lighting conditions and texture similarities between temporal images cause severe false alarms in competing methods. KGDNet mitigates

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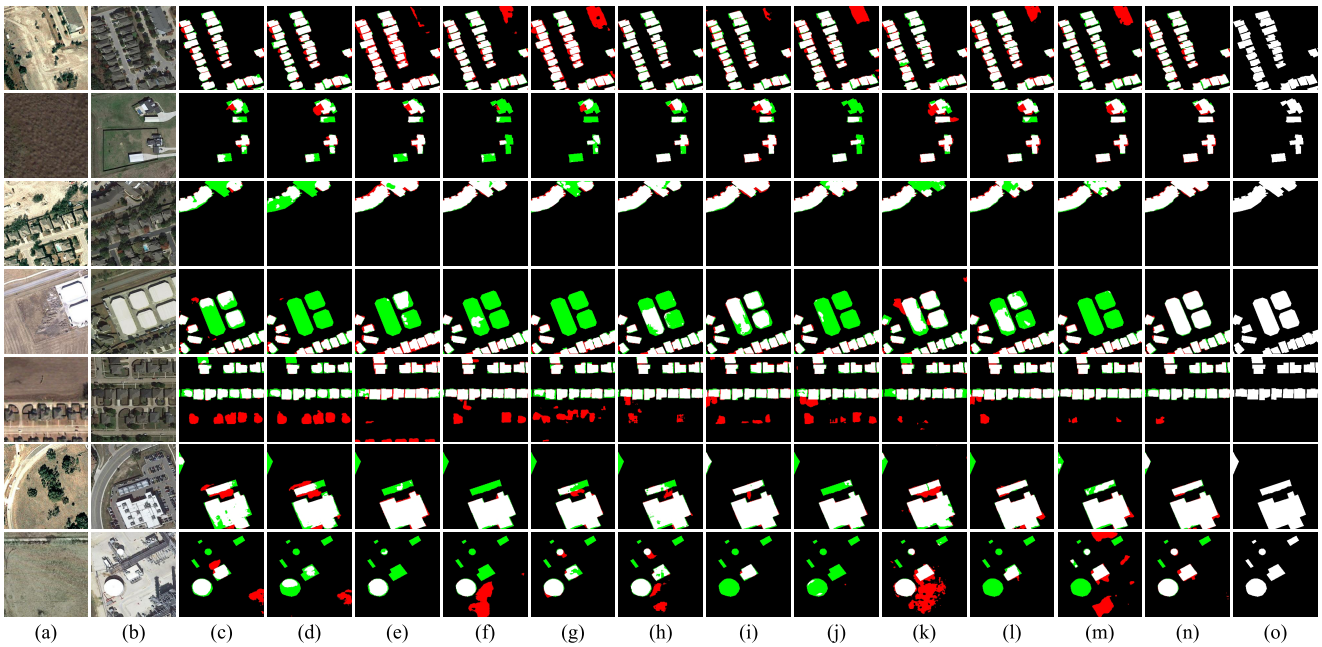


Fig. 8. Visualization results of different methods on the LEVIR+-CD test set. (a) T1 image. (b) T2 image. (c) FC-Siam-Conc. (d) FC-Siam-Diff. (e) STANet. (f) SNUNet. (g) BIT. (h) ChangeFormer. (i) AMTNet. (j) SEIFNet. (k) CDMamba. (l) SFEARNet. (m) CVNet. (n) KGDNet. (o) Ground truth. White denotes true positives, black represents true negatives, red indicates false positives, and green indicates false negatives.

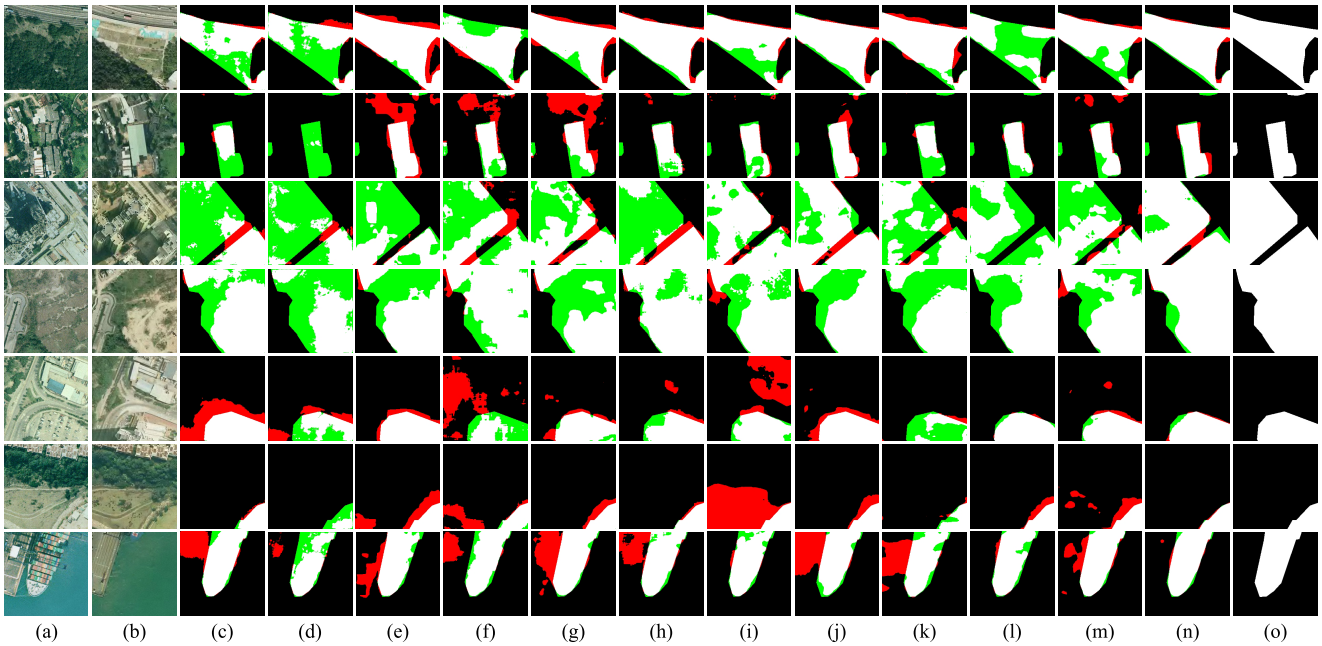


Fig. 9. Visualization results of different methods on the SYSU-CD test set. (a) T1 image. (b) T2 image. (c) FC-Siam-Conc. (d) FC-Siam-Diff. (e) STANet. (f) SNUNet. (g) BIT. (h) ChangeFormer. (i) AMTNet. (j) SEIFNet. (k) CDMamba. (l) SFEARNet. (m) CVNet. (n) KGDNet. (o) Ground truth. White denotes true positives, black represents true negatives, red indicates false positives, and green indicates false negatives.

these pseudo-changes by utilizing 3-D-aware representations that remain stable despite radiometric deviations.

Further validation on the SYSU-CD dataset (Fig. 9) demonstrates KGDNet's strong generalization across diverse land-cover changes. Row 1 involves subtle feature changes where most models exhibit missed detections and poor edge delineation; KGDNet, however, accurately captures this semantic shift. In urban scenarios, Row 3 shows that strong building shadows often lead to false positives or false negatives in other methods. Notably, Rows 4 and 6 depict the mutual transformations between vegetation and bare soil. Although these low-relief scenarios lack the significant elevation differences typically associated with human-made structures, KGDNet accurately identifies these semantic transformations, demonstrating its capability to capture changes beyond those with pronounced geometric variations. Finally, Row 7

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illustrates a complex port scenario where diverse container colors disrupt the spectral continuity of standard models, leading to fragmented results. KGDNet successfully integrates these diverse features to produce a complete and continuous detection of the port infrastructure.

Across all datasets, particularly in complex scenes containing dense and small-scale objects, KGDNet consistently preserves sharper boundaries than competing methods. Overall, the qualitative results are consistent with the quantitative evaluations, confirming that the proposed geometry-guided and mechanism-decoupled framework provides robust and accurate CD across diverse high-resolution remote sensing scenarios.

E. Ablation Study

To analyze the contribution of each key design in KGDNet, we conduct ablation studies to examine how geometry-guided representation learning and decoupled change modeling affect performance. Starting from a geometry-agnostic single-branch baseline, we progressively introduce prior injection, GAFM, MH-LSTA, and the tri-branch structure. In addition, multiscale ablation and efficiency comparisons are performed to analyze the impact of feature resolution and model efficiency.

1) Impact of Prior Knowledge: To rigorously evaluate the effectiveness and robustness of geometric priors in top-down remote sensing scenarios, we conducted a comprehensive ablation study by systematically comparing different MDE models and explicitly contrasting RGB-only versus RGB-D settings.

We first established an RGB-only baseline that relies solely on bitemporal optical images for CD. Subsequently, depth maps were injected as geometric priors, yielding RGB-D inputs. This comparison enables a direct evaluation of the contribution of geometric guidance, isolating the effect of priors from architectural factors.

To assess the out-of-domain generalization capability of monocular depth models in aerial scenes, five representative estimators were evaluated, including MiDaS, Depth Pro, and three variants of DAv2 (DAv2-S/B/L) with increasing model capacities. All depth maps were generated in a zero-shot manner without domain-specific fine-tuning, ensuring a fair assessment of cross-domain transferability.

Visual comparisons in Fig. 10 reveal that the structural reliability of depth priors is strongly correlated with model capacity. While larger models produce more coherent geometric layouts with clearer building boundaries, MiDaS and DAv2-S frequently exhibit semantic ambiguity, such as assigning erroneously high elevations to roads, vegetation, or shadowed regions, and failing to separate dense building clusters. In contrast, Depth Pro and DAv2-B provide improved structural consistency, whereas DAv2-L yields the sharpest object boundaries and the most stable responses under shadow and illumination variations. These observations confirm that depth estimation errors in remote sensing contexts are highly model-dependent and spatially structured rather than random.

Crucially, although artifacts are visible in the generated priors, they do not propagate into the final change predictions. As evidenced in Figs. 10 and 11, KGDNet effectively suppresses typical failure cases associated with imperfect depth estimation. For instance, in LEVIR-CD (Row 1), although trees in the upper left region trigger strong depth responses, the network correctly avoids predicting false changes. Similarly, in WHU-CD (Row 1), depth fluctuations on nonbuilding surfaces are accurately rejected. Furthermore, in LEVIR+-CD (Row 4) and SYSU-CD (Row 2), where depth estimators are misled by vegetation and shadowed structures, KGDNet maintains clean change maps without introducing corresponding false positives. These results demonstrate that KGDNet does not indiscriminately rely on geometric priors; instead, it is capable of selectively suppressing unreliable depth-induced activations, effectively filtering out geometric noise.

TABLE III

QUANTITATIVE COMPARISON (%) OF DIFFERENT MDE MODELS USED AS GEOMETRIC PRIORS. RGB DENOTES MODELS TRAINED WITH OPTICAL IMAGES ONLY, WHILE RGB-D DENOTES MODELS TRAINED WITH ADDITIONAL DEPTH PRIORS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Type	Models	OA	F1	IoU	Rec
LEVIR-CD	RGB	-	99.11	90.65	83.07	90.67
	RGB-D	MiDaS	99.08	90.63	82.90	90.21
	RGB-D	Depth Pro	99.16	91.79	84.37	**91.58**
	RGB-D	DAv2-S	99.07	90.25	82.29	90.03
	RGB-D	DAv2-B	99.15	91.56	84.15	91.01
	RGB-D	DAv2-L	**99.23**	**92.30**	**85.71**	91.12
WHU-CD	RGB	-	99.41	92.72	86.78	92.31
	RGB-D	MiDaS	99.43	93.04	87.11	92.03
	RGB-D	Depth Pro	99.50	93.89	88.42	93.40
	RGB-D	DAv2-S	99.43	93.11	87.24	92.44
	RGB-D	DAv2-B	99.49	93.77	88.32	92.85
	RGB-D	DAv2-L	**99.54**	**94.71**	**89.96**	**93.44**
LEVIR+-CD	RGB	-	98.61	82.19	70.05	81.36
	RGB-D	MiDaS	98.58	81.61	69.33	80.42
	RGB-D	Depth Pro	98.76	84.44	73.33	83.91
	RGB-D	DAv2-S	98.67	83.09	71.38	83.23
	RGB-D	DAv2-B	98.76	84.20	72.99	84.14
	RGB-D	DAv2-L	**98.79**	**85.10**	**74.06**	**84.81**
SYSU-CD	RGB	-	90.72	80.18	67.00	80.31
	RGB-D	MiDaS	90.80	79.66	66.28	79.63
	RGB-D	Depth Pro	91.34	81.66	69.12	82.01
	RGB-D	DAv2-S	90.57	80.11	66.85	79.68
	RGB-D	DAv2-B	91.26	81.18	68.35	80.11
	RGB-D	DAv2-L	**91.92**	**83.05**	**71.01**	**83.91**

The quantitative results in Table III further reveal that detection performance depends on both prior quality and the network's ability to regulate prior influence, while the sensitivity to prior reliability varies across datasets. On LEVIR-CD, low-capacity depth models tend to introduce misleading geometric cues by misclassifying roads and ground textures as elevated structures; consequently, RGB-D models using MiDaS and DAv2-S perform slightly worse than the RGB-only baseline, indicating that naive prior injection can be detrimental when geometric noise dominates. As prior quality improves, detection performance increases consistently, and DAv2-L achieves the best results. On WHU-CD, building structures are salient and scene variability is relatively

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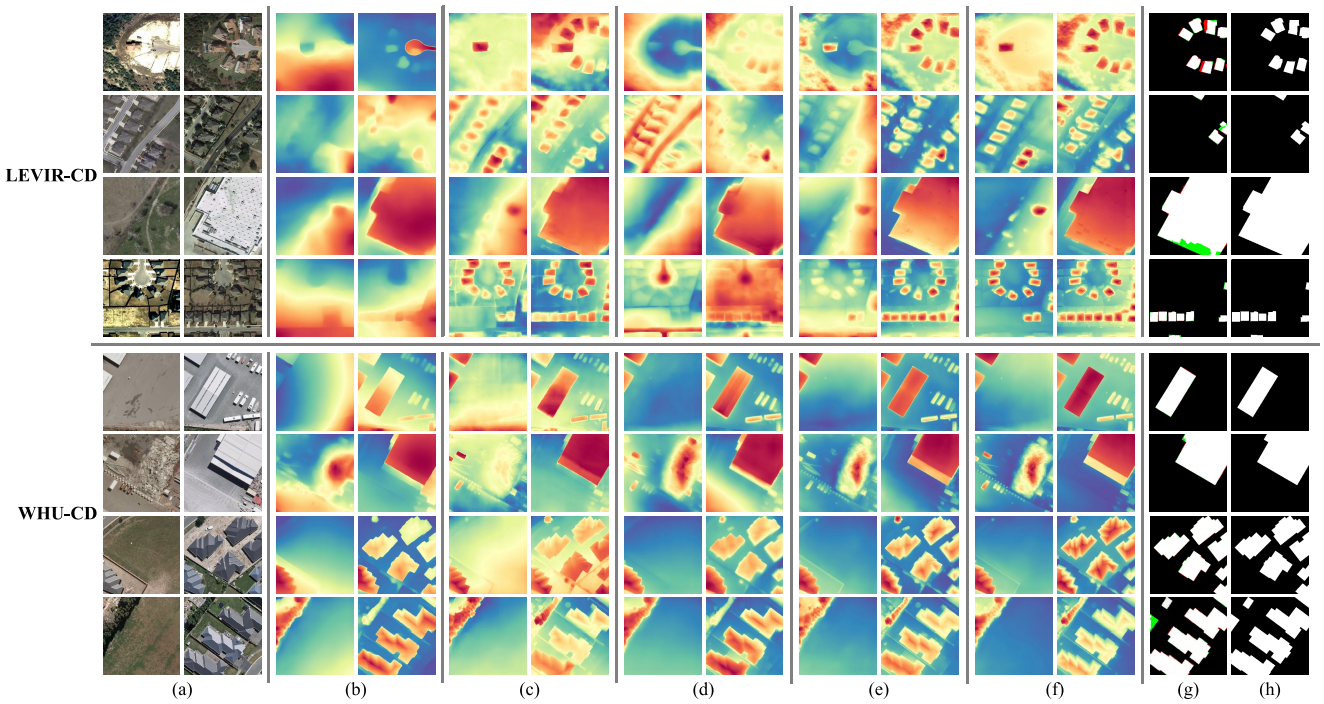


Fig. 10. Visualization results of prior knowledge generated by different depth estimation models on the LEVIR-CD and WHU-CD datasets. (a) Bitemporal image pair. (b) MiDaS. (c) Depth Pro. (d) DAv2-S. (e) DAv2-B. (f) DAv2-L. (g) KGDNet. (h) Ground truth. Red indicates higher elevation values, while blue indicates lower elevation values.

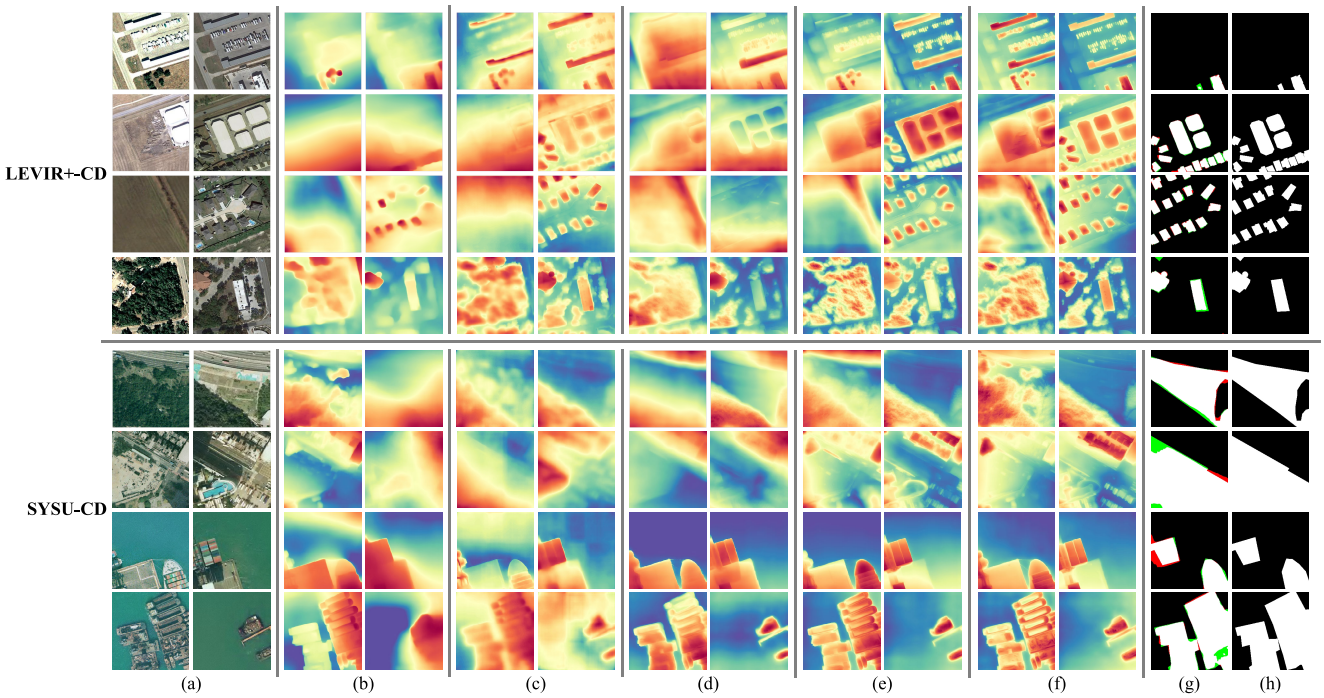


Fig. 11. Visualization results of prior knowledge generated by different depth estimation models on the LEVIR+-CD and SYSU-CD dataset. (a) Bitemporal image pair. (b) MiDaS. (c) Depth Pro. (d) DAv2-S. (e) DAv2-B. (f) DAv2-L. (g) KGDNet. (h) Ground truth. Red indicates higher elevation values, while blue indicates lower elevation values.

limited; under such conditions, even coarse geometric cues help reinforce object localization, enabling all depth models to outperform the RGB-only baseline, while high-quality priors still lead to further gains, with DAv2-L yielding the highest F1. In the more challenging LEVIR+-CD dataset, which contains complex roof materials and dense vegetation, low-quality priors provide unstable or marginal benefits, whereas Depth Pro and all DAv2 variants deliver consistent improvements, and performance increases monotonically from DAv2-S to DAv2-L, with the highest capacity model

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TABLE IV

QUANTITATIVE COMPARISON (%) OF GAFM WITH AND WITHOUT CROSS-MODAL AND CROSS-TEMPORAL ALIGNMENT. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	GAFM-K	GAFM-T	OA	F1	IoU	Rec
LEVIR-CD	×	×	99.01	89.85	81.40	89.71
	✓	×	99.14	91.29	83.96	90.68
	×	✓	99.11	90.73	83.09	90.52
	✓	✓	**99.23**	**92.30**	**85.71**	**91.12**
WHU-CD	×	×	99.31	91.48	84.46	91.61
	✓	×	99.43	93.12	87.22	92.21
	×	✓	99.36	92.44	86.07	92.40
	✓	✓	**99.54**	**94.71**	**89.96**	**93.44**
LEVIR+-CD	×	×	98.32	79.71	66.65	80.07
	✓	×	98.74	84.15	72.90	**84.97**
	×	✓	98.69	83.48	71.96	84.02
	✓	✓	**98.79**	**85.10**	**74.06**	84.81
SYSU-CD	×	×	91.05	80.29	67.21	79.76
	✓	×	91.34	81.47	68.76	79.97
	×	✓	91.35	81.27	68.48	80.10
	✓	✓	**91.92**	**83.05**	**71.01**	**83.91**

TABLE V

QUANTITATIVE COMPARISON (%) OF GAFM AND OTHER ATTENTION MECHANISMS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Method	OA	F1	IoU	Rec
LEVIR-CD	Baseline	99.01	89.85	81.40	89.71
	+ SE-Block [67]	98.98	89.89	81.63	88.57
	+ CBAM [68]	98.98	89.82	81.53	88.78
	+ GAFM (Ours)	**99.23**	**92.30**	**85.71**	**91.12**
WHU-CD	Baseline	99.31	91.48	84.46	91.61
	+ SE-Block [67]	99.38	92.68	86.35	89.35
	+ CBAM [68]	99.19	90.41	82.50	86.79
	+ GAFM (Ours)	**99.54**	**94.71**	**89.96**	**93.44**
LEVIR+-CD	Baseline	98.32	79.71	66.65	80.07
	+ SE-Block [67]	98.45	81.02	68.10	81.16
	+ CBAM [68]	98.54	81.58	68.89	79.36
	+ GAFM (Ours)	**98.79**	**85.10**	**74.06**	**84.81**
SYSU-CD	Baseline	91.05	80.29	67.21	79.76
	+ SE-Block [67]	90.35	78.17	64.16	73.28
	+ CBAM [68]	89.39	75.89	61.14	70.81
	+ GAFM (Ours)	**91.92**	**83.05**	**71.01**	**83.91**

improving F1 by 2.91% over the RGB baseline. Finally, on SYSU-CD, which involves diverse change categories, oblique viewing angles, and pronounced building shadows, low-quality priors offer only limited or unstable improvements, while high-quality geometric cues substantially enhance robustness against shadow-induced pseudo-changes and improve boundary completeness, allowing DAv2-L to outperform the RGB baseline by 2.87% in $F1$.

Overall, detection performance exhibits a consistent upward trend as depth quality improves from low-capacity to high-capacity models across benchmarks with varying scene complexities. However, the qualitative results indicate that KGDNet can suppress localized depth errors even when imperfect priors are provided, preventing them from being directly translated into false change predictions.

This observation highlights that geometric priors should not be treated as static and fully reliable signals, but rather as auxiliary cues that require adaptive regulation. Consequently, KGDNet incorporates a dedicated difference branch together with DPM, which dynamically generates confidence maps from geometry-aware cross-temporal differences. These confidence maps downweight unreliable prior-induced activations in the appearance and disappearance branches, effectively preventing error accumulation from imperfect depth estimation. Based on both quantitative performance and qualitative stability, DAv2-L is adopted as the default prior generator in all subsequent experiments.

2) Impact of GAFM: To evaluate the effectiveness of GAFM, we conducted extensive ablation studies on four benchmarks: LEVIR-CD, WHU-CD, LEVIR+-CD, and SYSU-CD. As detailed in Table IV, progressively enabling the cross-modal alignment (GAFM-K) and cross-temporal fusion (GAFM-T) components yields consistent performance improvements. While individual activation of either component provides stable gains, their joint deployment delivers the most significant boost, indicating that geometric guidance and temporal interaction are complementary. This advantage is particularly pronounced on the challenging LEVIR+-CD dataset, where the full configuration increases the $F1$ by 5.39%. Even on SYSU-CD, which involves diverse land-cover changes, the proposed module achieves a robust 2.76% improvement, validating its efficacy in handling complex and heterogeneous change patterns.

To further validate the superiority of the proposed guidance-driven design, GAFM is compared with representative self-attention mechanisms, specifically SE-Block and CBAM. Unlike these conventional modules that rely on self-modulation derived solely from current features, GAFM explicitly incorporates external geometric and temporal priors. This structural distinction leads to divergent performance in complex scenarios. As shown in Table V, while SE-Block and CBAM provide marginal gains on standard building datasets, they struggle to generalize on the diverse SYSU-CD, resulting in $F1$ drops of 2.12% and 4.40%, respectively. This suggests that without structural constraints, traditional attention is prone to overfitting spectral noise. Conversely, GAFM maintains a consistent improvement across all benchmarks, highlighting its superior adaptability and robustness.

Qualitative comparisons in Fig. 12 corroborate these quantitative findings. Lacking explicit geometric guidance, standard attention mechanisms tend to generate response maps with blurred boundaries and are susceptible to background interference. In contrast, GAFM leverages structural priors to yield sharper activations with clearer contours, effectively suppressing spurious background noise. Moreover, in challenging cases involving atypical appearances or weak spectral contrast where traditional modules often produce fragmented

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TABLE VI
QUANTITATIVE COMPARISON (%) OF EFFICIENCY AND ACCURACY FOR DIFFERENT ATTENTION MECHANISMS AND MH-LSTA. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

	Efficiency			LEVIR-CD (%)		WHU-CD (%)		LEVIR+-CD (%)		SYSU-CD (%)	
Attention	Params (M)	FLOPs (G)	Time (ms)	F1	IoU	F1	IoU	F1	IoU	F1	IoU
Global	11.388	16.211	21.979	91.66	84.61	94.07	88.88	84.21	73.02	82.06	69.57
Window	11.388	4.131	12.939	91.11	83.69	93.45	87.87	82.82	70.99	81.29	68.48
Linear	11.388	3.896	10.798	90.71	83.04	93.56	87.93	81.42	69.05	80.60	67.65
MH-LSTA	7.122	3.304	10.621	**92.30**	**85.71**	**94.71**	**89.96**	**85.10**	**74.06**	**83.05**	**71.01**

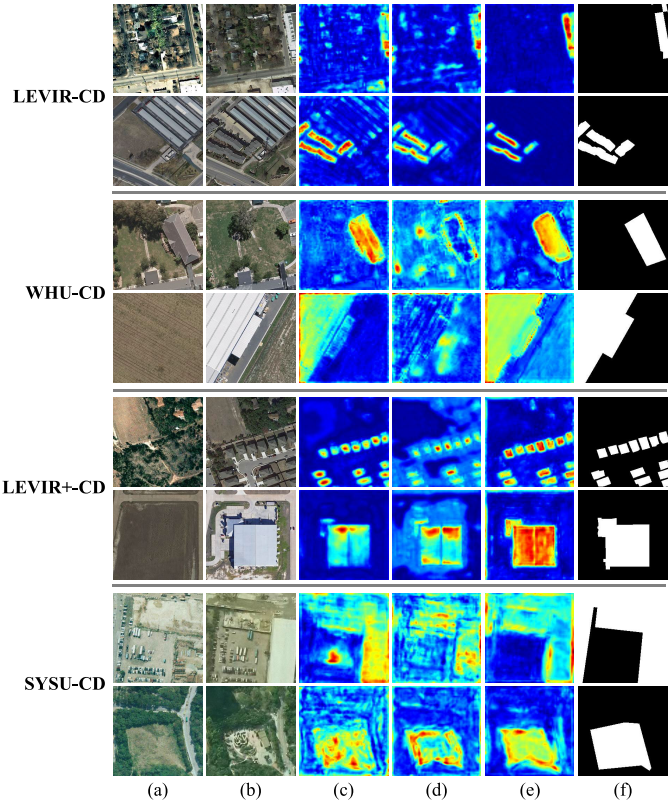


Fig. 12. Visualization comparison of different fusion modules for validating the effectiveness of GAFM. (a) T1 image. (b) T2 image. (c) SE-Block. (d) CBAM. (e) GAFM. (f) Ground truth. Red indicates a higher response to changed regions, while blue indicates a lower response.

responses, GAFM maintains feature continuity and captures change regions in a more complete and accurate manner.

Based on both quantitative and qualitative evidence, the joint GAFM-K and GAFM-T configuration is adopted as the default setting of the proposed KGDNet.

3) Impact of MH-LSTA: To evaluate both the effectiveness and efficiency of the proposed MH-LSTA, we compare it with several representative attention mechanisms, including global, window-based, and linear attention. The results are summarized in Table VI. For a fair comparison, all methods are evaluated using the same network and the reported parameters, FLOPs, and inference time are measured for the entire CD network with an input resolution of 256×256.

Compared with standard global attention, MH-LSTA significantly reduces computational cost while achieving superior detection accuracy across all datasets. By performing attention in a compressed latent space with an asymmetric design, it achieves the lowest FLOPs and inference time among all compared methods and also uses fewer parameters at the network level. Meanwhile, it consistently delivers the best $F1$ and IoU scores, indicating that the efficiency gain does not come at the expense of accuracy. This suggests that latent-space attention with asymmetric interaction is well suited for high-resolution CD, where dense self-attention is computationally prohibitive and often redundant.

We further analyze the impact of the latent compression ratio r and Geo-PE, as reported in Table VII. In this experiment, FLOPs and parameter counts are measured for the MH-LSTA module itself. Increasing r steadily reduces FLOPs and parameters, validating the effectiveness of the latent compression strategy. Notably, moderate compression not only improves efficiency but also leads to better accuracy, indicating that compressing high-dimensional features can effectively suppress redundant responses and enhance the reliability of semantic representations. In contrast, overly aggressive compression causes clear performance degradation, suggesting that excessive dimensionality reduction discards critical discriminative cues.

Importantly, although key and value features are compressed, query features preserve the original spatial resolution, which allows local structural details to be retained during cross-feature interaction. This asymmetric compression strategy enables MH-LSTA to balance global context modeling and local detail preservation, making it particularly suitable for fine-grained change localization.

In addition, enabling Geo-PE consistently improves performance across all compression ratios with negligible computational overhead. This confirms that explicit spatial encoding effectively complements latent-space attention by enhancing geometric and positional awareness, which is critical for accurate change localization.

As further illustrated in Fig. 13, the computational advantage of MH-LSTA becomes more pronounced as input resolution increases, making it particularly suitable for large-scale and high-resolution remote sensing imagery. Based on this analysis, we adopt $r = 4$ with Geo-PE enabled as the default configuration in all experiments.

4) Impact of Tri-Branch Decoupled Structure: To verify the mechanism-level interpretability and effectiveness of the proposed tri-branch decoupled architecture, we conduct a systematic ablation study by removing each specialized

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TABLE VII

QUANTITATIVE COMPARISON (%) OF MH-LSTA UNDER DIFFERENT COMPRESSION RATIOS AND GEO-PE SETTINGS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Ratio	Geo-PE	FLOPs (G)	Params (M)	LEVIR-CD (%)		WHU-CD (%)		LEVIR+-CD (%)		SYSU-CD (%)	
				F1	IoU	F1	IoU	F1	IoU	F1	IoU
1	×	2.477	5.694	91.61	84.54	92.76	86.61	83.78	72.09	81.55	68.84
	✓	2.714		91.84	84.72	93.62	87.97	83.92	72.61	81.77	69.40
2	×	1.770	4.272	91.07	84.12	93.17	87.29	83.61	72.12	81.98	69.54
	✓	2.006		91.71	84.34	94.25	89.12	83.85	72.52	82.15	69.71
4	×	1.416	3.561	91.51	84.98	93.34	87.44	84.06	72.75	82.04	69.84
	✓	1.652		**92.30**	**85.71**	**94.71**	**89.96**	**85.10**	**74.06**	**83.05**	**71.01**
8	×	1.239	3.206	90.66	82.98	91.63	86.18	81.28	68.88	79.66	66.27
	✓	1.475		91.04	83.74	92.22	86.94	82.68	70.85	80.83	67.86

TABLE VIII

QUANTITATIVE COMPARISON (%) OF DIFFERENT BRANCH COMBINATIONS IN THE TRI-BRANCH DECOUPLED STRUCTURE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Branch A	Branch B	Branch C	OA	F1	IoU	Rec
LEVIR-CD	×	✓	✓	99.13	91.37	84.11	90.39
	✓	×	✓	99.07	90.36	82.49	90.26
	✓	✓	×	99.10	91.05	83.57	90.27
	✓	✓	✓	**99.23**	**92.30**	**85.71**	**91.12**
WHU-CD	×	✓	✓	99.42	93.77	88.27	93.56
	✓	×	✓	99.34	92.41	86.19	92.73
	✓	✓	×	99.41	93.91	88.63	**94.03**
	✓	✓	✓	**99.54**	**94.71**	**89.96**	93.44
LEVIR+-CD	×	✓	✓	98.69	83.76	72.31	84.04
	✓	×	✓	98.59	82.53	70.54	82.47
	✓	✓	×	98.68	83.19	71.48	83.71
	✓	✓	✓	**98.79**	**85.10**	**74.06**	**84.81**
SYSU-CD	×	✓	✓	91.35	81.73	69.34	81.14
	✓	×	✓	91.05	80.29	67.21	79.65
	✓	✓	×	91.39	81.11	68.33	80.97
	✓	✓	✓	**91.92**	**83.05**	**71.01**	**83.91**

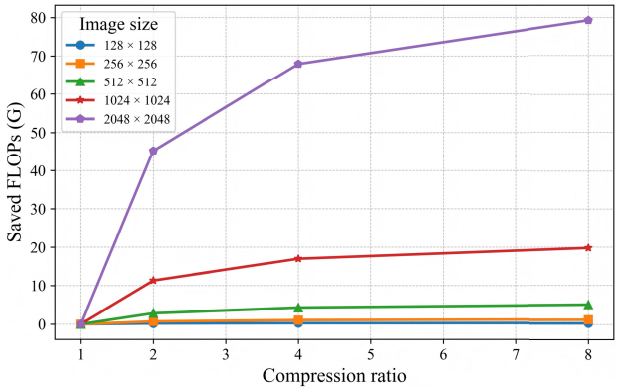


Fig. 13. Saved FLOPs (G) in the MH-LSTA module as a function of compression ratio (r) for different input image sizes.

branch: Branches A–C. The quantitative results are reported in Table VIII, and qualitative mechanism-specific responses are visualized in Fig. 14. As shown in Table VIII, removing any single branch consistently degrades performance across all benchmarks, while the full tri-branch model achieves the best OA, $F1$, and IoU, confirming that the three branches provide complementary and nonredundant contributions to change modeling.

Removing Branch A mainly weakens the detection of newly emerging objects. For example, on LEVIR-CD, $F1$ drops from 92.30% to 91.37%, and IoU decreases from 85.71% to 84.11%. On LEVIR+-CD, F1 further decreases from 85.10% to 83.76%, indicating that forward appearance modeling is particularly important for identifying new constructions in complex urban scenes.

Removing Branch C primarily affects the detection of disappearing structures. On LEVIR-CD, IoU decreases from 85.71% to 83.57%, and on SYSU-CD, it drops from 71.01% to 68.33%, demonstrating that disappearance patterns cannot be reliably inferred from appearance or generic difference cues alone, especially in diverse urban environments.

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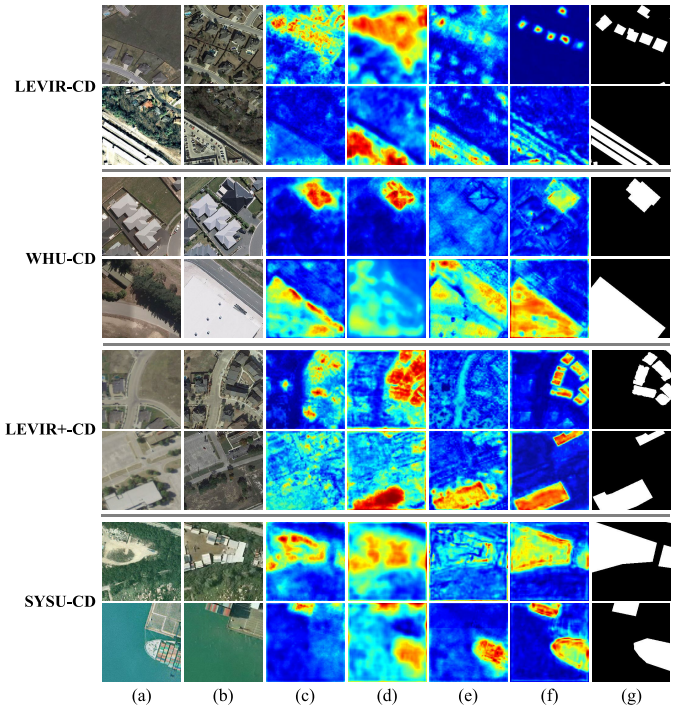


Fig. 14. Visualization comparison of mechanism-specific responses of the tri-branch architecture for validating the interpretability of decoupled change modeling. (a) T1 image. (b) T2 image. (c) Branch A (appearance) exclusively highlights newly emerging objects. (d) Branch B (difference) functions as a geometric modulator to refine boundaries and suppress noise. (e) Branch C (disappearance) conversely targets vanished structures. (f) Fusion result of the three branches integrates these mechanism-aware cues to yield precise detection. (g) Ground truth. Red denotes higher confidence of change, while blue denotes lower confidence.

TABLE IX
QUANTITATIVE COMPARISON (%) OF DIFFERENT MULTISCALE FEATURE COMBINATIONS (S1–S4). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	S1	S2	S3	S4	OA	F1	IoU	Rec
LEVIR-CD	✓	×	×	×	99.05	89.99	81.88	90.03
	✓	✓	×	×	99.10	90.75	83.13	90.81
	✓	✓	✓	×	99.14	91.81	84.87	**91.17**
	✓	✓	✓	✓	**99.23**	**92.30**	**85.71**	91.12
WHU-CD	✓	×	×	×	99.24	91.04	83.56	91.73
	✓	✓	×	×	99.35	92.22	85.72	92.81
	✓	✓	✓	×	99.43	93.26	87.45	93.15
	✓	✓	✓	✓	**99.54**	**94.71**	**89.96**	**93.44**
LEVIR+-CD	✓	×	×	×	98.61	82.43	70.38	82.15
	✓	✓	×	×	98.68	83.79	72.37	83.51
	✓	✓	✓	×	98.73	84.09	72.79	83.73
	✓	✓	✓	✓	**98.79**	**85.10**	**74.06**	**84.81**
SYSU-CD	✓	×	×	×	91.10	80.69	67.85	79.45
	✓	✓	×	×	91.38	81.22	68.48	81.09
	✓	✓	✓	×	91.53	82.24	69.84	81.80
	✓	✓	✓	✓	**91.92**	**83.05**	**71.01**	**83.91**

The most significant performance degradation is observed when removing Branch B, which is constructed by explicitly computing the absolute difference between the two temporally fused feature maps, followed by an independent MSD. On WHU-CD, IoU drops from 89.96% to 86.19%, and on LEVIR+-CD, it decreases from 74.06% to 70.54%, indicating that without an explicit difference pathway, the network becomes more vulnerable to pseudo-changes and ambiguous temporal variations.

Beyond numerical performance, Fig. 14 provides direct evidence of mechanism-specific responses learned by each branch. Since no explicit differencing operation is introduced before the tri-branch stage, both Branches A and C exhibit noticeable activation over changed regions in the heatmaps, but in a directionally asymmetric manner, Fig. 14(c) shows that Branch A exhibits a significantly stronger response to newly emerging objects, while Fig. 14(e) demonstrates that Branch C responds more intensely to disappearing structures. This demonstrates that the two branches learn complementary temporal semantics rather than redundant change signals.

In contrast, Branch B, derived from explicit feature differencing, acts as a selector of change candidates by strengthening responses in truly changed areas while suppressing activations in unchanged regions. The decoded difference feature is further processed by DPM to generate a spatial confidence map. As visualized in Fig. 14(d), this map serves as a geometric modulator that gates the outputs of Branches A and C, preventing spurious activations from propagating to the final prediction.

After fusion, the gated appearance and disappearance features are integrated into a unified change representation, as shown in Fig. 14(f), yielding predictions that align closely with the ground truth. Overall, both quantitative ablation and qualitative visualization verify that explicitly decoupling appearance, difference, and disappearance mechanisms improves detection accuracy while enabling interpretable, mechanism-aware change modeling.

5) Impact of Multiscale Features: To evaluate the effectiveness of the multiscale fusion strategy in the MSD, we progressively integrate feature maps from different scales, as reported in Table IX. Starting from the finest scale S1, coarser and more semantic scales (S2–S4) are gradually added.

Across all datasets, performance consistently improves as more scales are fused, demonstrating the necessity of hierarchical feature aggregation. Using only S1 yields the lowest accuracy since fine-scale features lack sufficient semantic context. Introducing deeper scales steadily boosts F1 and IoU, indicating that coarse-grained semantic information effectively complements fine-grained spatial details.

With all four scales enabled, the model achieves the best overall performance on all benchmarks, especially on WHU-CD and LEVIR+-CD, where deeper features contribute significantly to detecting large structures and subtle changes, respectively. These results confirm that combining multilevel semantic and spatial features is critical for robust and accurate CD.

6) Comparison of Model Efficiency: Beyond accuracy, computational complexity is pivotal for practical RSCD. As reported in Table X, KGDNet achieves a competitive balance between performance and cost. The model maintains

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TABLE X
QUANTITATIVE COMPARISON OF EFFICIENCY AMONG DIFFERENT CD METHODS

Methods	Params (M)	FLOPs (G)
FC-Siam-Conc [19]	1.55	5.33
FC-Siam-Diff [19]	1.35	4.73
STANet [69]	16.93	13.16
SNUNet [29]	12.03	54.83
BIT [20]	3.04	8.75
ChangeFormer [32]	41.03	202.79
AMTNet [38]	24.67	21.56
SEIFNet [30]	27.91	8.37
ChangeDA [62]	30.25	23.49
CDMamba [43]	8.88	54.18
SFEARNet [14]	5.56	4.65
CVNet [60]	88.79	86.60
KGDNet (Ours)	26.94	12.12

a moderate parameter scale, which is mainly attributed to the tri-branch decoupled structure where independent weights are allocated to appearance, disappearance, and difference branches to preserve mechanism-specific representations.

Importantly, the parameter scale does not lead to proportional computational overhead. KGDNet requires lower FLOPs than most compared methods in Table X. This efficiency stems from two key designs: 1) GAFM is applied only at the knowledge embedding and cross-temporal fusion stages, rather than repeatedly in the decoder; and 2) the embedded MH-LSTA block performs asymmetric interaction in a compressed latent space, projecting keys and values into low dimensions while preserving full-resolution queries.

As a result, the model achieves relatively low floating-point operations while preserving detection accuracy. Compared with Transformer-based approaches that rely on dense self-attention on high-resolution features, KGDNet concentrates computation on compact latent representations, providing a practical and computation-efficient alternative for large-scale remote sensing applications.

V. DISCUSSION

The superior performance of KGDNet is fundamentally attributed to the synergistic interplay among three core architectural innovations, which collectively provide a robust solution to key challenges in RSCD.

First, KGDNet upgrades the modeling space from a 2-D feature comparison to 3-D structural awareness through knowledge guidance. The framework introduces geometric priors, and our GAFM module efficiently leverages this knowledge to inherently suppress pseudo-changes, such as shadows and illumination variations, which lack genuine structural shifts. Crucially, the difference branch performs a vital dual role: beyond learning temporal differences, it acts as a dynamic refinement mechanism to calibrate the prior, ensuring robustness against the high sensitivity to low-quality depth cues—a pervasive issue in knowledge-guided approaches.

Second, we challenge the conventional unified binary supervision paradigm by explicitly decoupling the CD task into appearance, difference, and disappearance branches. This synergistic tri-branch structure achieves a more balanced and comprehensive perception of emerging and vanishing objects. By overcoming the systematic biases of monolithic models, this design facilitates a more interpretable and robust understanding of heterogeneous change mechanisms.

Third, KGDNet tackles the computational bottleneck of high-resolution processing through the efficient MH-LSTA module. Our findings demonstrate that moderate feature compression acts as an effective regularizer—not only drastically reducing the computational load but also yielding concurrent gains in accuracy. This design successfully strikes an optimal balance between the effectiveness of global dependency modeling and practical computational scalability.

Despite its strong results, KGDNet has two main limitations. The first is its reliance on external MDE. Although depth priors are annotation-free, they are not jointly optimized with the RSCD objective and can exhibit systematic errors over vegetation, water, or low-texture areas. DPM and difference branch mitigate this issue by dynamically downweighting unreliable regions, but failure cases still arise where severely biased depth maps introduce artifacts or suppress subtle real changes. The second limitation is that, although the internal representation is mechanism-decoupled, the final output remains a binary change map. Collapsing rich mechanism-specific features into a single channel forfeits a valuable opportunity for interpretability and semantic analysis of change types.

In future work, these observations suggest several directions. On the prior side, integrating uncertainty-aware or jointly fine-tuned depth estimation, or exploring alternative priors such as DSMs and learned geometric embeddings, may further stabilize the guidance. On the output side, leveraging the tri-branch feature spaces to predict multiclass or semantic change maps (e.g., building expansion versus demolition) would extend the paradigm from "CD" toward "change understanding," aligning better with practical demands in urban monitoring and land management.

VI. CONCLUSION

To address the core challenges in high-resolution RSCD—namely, prevalent pseudo-change interference, high computational overhead, and insufficient modeling of diverse change mechanisms—this article proposes a novel, knowledge-guided, tri-branch network, KGDNet, that decouples the modeling of change mechanisms. By leveraging annotation-free monocular depth maps as geometric prior knowledge, the network upgrades traditional 2-D appearance features into 3-D, geometry-aware hybrid representations, thereby significantly enhancing its ability to discriminate between genuine land-cover changes and pseudo-changes.

The core innovation of KGDNet is embodied in three synergistically designed modules. First, the knowledge-guided feature encoder uses high-quality depth maps as guidance to provide stable constraints at the pixel, geometric, and structural levels, effectively suppressing pseudo-changes. Second, the MH-LSTA module performs feature compression and remapping in a latent space, substantially reducing computational and memory costs while strengthening geometric-structure

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perception and spatiotemporal relationship modeling; this achieves a favorable balance between accuracy and efficiency. Finally, the tri-branch, decoupled change modeling component is, to the best of our knowledge, one of the first in RSCD to explicitly decouple the three core change mechanisms; it learns heterogeneous change patterns through specialized branches and leverages the difference branch to dynamically refine the prior knowledge.

Experiments on multiple public benchmarks demonstrate that KGDNet achieves the best performance among the compared methods—including CNN-, Transformer-, and Mamba-based methods—across multiple metrics, showcasing its accuracy, robustness, and generalization capability. This study not only provides a strong new baseline for high-resolution RSCD but also offers, through its design philosophy of knowledge guidance and mechanism decoupling, a new paradigm for addressing complex problems in the field.

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